$SL_3(\mathbb{R})$ REPRESENTATIONS OF IDEAL TRIANGLE GROUPS FROM THE INSIDE

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Introduction

Any three distinct points on the boundary of the hyperbolic plane span a unique ideal triangle, and any two ideal triangles are isometric. Gluing these rigid pieces with a shear on each side gives coordinates on the Teichmüller space of finite-area hyperbolic surfaces [Bes+13], and even on the universal Teichmüller space [ŠWW24].

This picture has been very influential in higher rank. Fock–Goncharov parametrized convex projective structures on punctured surfaces by assigning positive configurations of flags to the punctures [FG07], and these coordinates were later extended to closed surfaces and to Hitchin components of higher rank [FG04; BD14]. While these coordinates help study Hitchin components, their connection to the geometry of the symmetric space is unclear.

In this paper, we define an analog of the ideal triangle for $G = SL_3(\mathbb{R})$. Given a positive triple of flags (F_1, F_2, F_3), we associate a domain $\mathcal{D}(F_1, F_2, F_3)$ in the symmetric space $\mathbb{X}$, bounded by three hypersurfaces that are asymptotic to the three flags, in an appropriate sense. When the flags are tangent to an ellipse in $\mathbb{RP}^2$, the domain coincides with the preimage of an ideal triangle under the closest point projection onto a totally geodesic hyperbolic plane.

The main difficulty in finding such a domain is that the symmetric space has no totally geodesic hypersurfaces, so it has no natural analog of the sides of an ideal triangle. Our hypersurfaces, which we call *walls*, are thus not totally geodesic, but they are foliated by totally geodesic submanifolds called *parallel sets*. This is reminiscent of bisectors in complex hyperbolic geometry, which are foliated by complex geodesics [Gol99].

As an application of the domains, we study convex projective structures on the orbifold associated to the group generated by reflections along the sides of an ideal triangle of $\mathbb{H}^2$, which we denote by W_3. The holonomy of these convex projective structures coincides with the relative Hitchin representations of W_3 into $SL_3(\mathbb{R})$. Positive triples of flags parametrize these representations.

The setup. An irreducible hyperbolic slice $\mathcal{H}$ in the symmetric space $\mathbb{X}$ is a totally geodesic copy of $\mathbb{H}^2$ whose stabilizer in $SL_3(\mathbb{R})$ is $SO(B)$, for B some $(1,2)$ form in $\mathbb{R}^3$. A key observation is that the closest-point projection $\pi : \mathbb{X} \to \mathcal{H}$ defines a fibration of $\mathbb{X}$, with totally geodesic fibers called *parallel sets*.

Parallel sets are isometric to $\mathbb{H} \times \mathbb{R}$, where $\mathbb{H}$ is a totally geodesic copy of the hyperbolic plane stabilized by a reducible embedding of $SL_2(\mathbb{R})$ into $SL_3(\mathbb{R})$. The space of parallel sets is parametrized by the homogeneous space

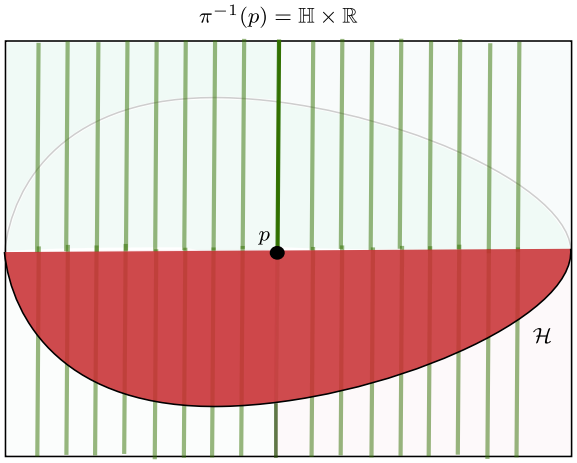


FIGURE 1. Schematic of a wall. The red region is an irreducible slice, and the wall divides $\mathbb{X}$ into two regions.

$$\mathbb{Y} = \{(p, \ell) \in \mathbb{RP}^2 \times (\mathbb{RP}^2)^* : p \notin \ell\}.$$

This space has plenty of structure; for instance, it possesses a natural symplectic form as described in [Lab07]. We will focus on its structure as a pseudo-Riemannian symmetric space. Crucially, two parallel sets joined by a spacelike geodesic are disjoint and their distance in $\mathbb{X}$ can be computed using the cross ratio (see Proposition 2.8).

Given a geodesic γ in $\mathcal{H}$, $\pi^{-1}(\gamma)$ is the union of the parallel sets corresponding to a spacelike geodesic in $\mathbb{Y}$. Conversely, any spacelike geodesic in $\mathbb{Y}$ is of this form. The union of these parallel sets is a 4-dimensional hypersurface of $\mathbb{X}$ which we call a *wall*.

Spacelike geodesics in $\mathbb{Y}$ are asymptotic to a pair of flags in general position. Conversely, given a pair of flags in general position, there are two spacelike geodesics asymptotic to them (see Proposition 2.12).

Triples of flags in general position come in two kinds, positive and negative (see Figure 2). We prove that, given a positive triple of flags, there is a canonical choice of spacelike geodesics for each pair. We prove that the walls associated to these three geodesics are disjoint and bound an open set in $\mathbb{X}$ which we call the *parallel domain* associated to the triple.

When the three flags are tangent to an ellipse, the parallel domain is the preimage of an ideal triangle under the closest-point projection onto an irreducible hyperbolic slice.

A useful feature of the parallel domain is that, using cross ratios, we can explicitly compute distances between parallel sets contained in walls in the boundary of the domain (see Proposition 3.5).

Main results. We apply the ideas explained above to study representations of the group W_3. A presentation for this group is

$$W_3 = \left\langle s_1, s_2, s_3 \mid s_i^2 \right\rangle.$$

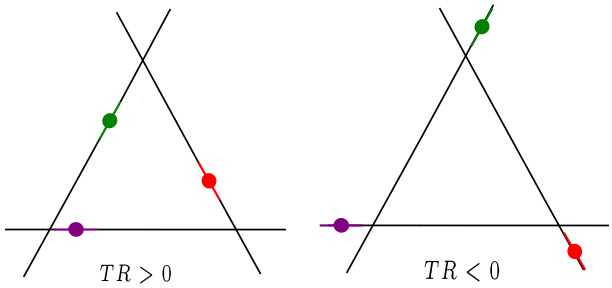


FIGURE 2. The triple ratio is a projective invariant of a triple of flags. The figure depicts a positive and a negative triplet.

More specifically, we study representations of this group into $SL_3(\mathbb{R})$ in which the products of pairs of generators are unipotent and fix a unique flag. These representations are parametrized by triples of flags fixed by these three products. With this in mind, we prove the following theorem:

Theorem 1 (Theorem 4.2)**.** Let (F_1, F_2, F_3) be a positive triple of flags, $\rho : W_3 \to SL_3(\mathbb{R})$ the associated representation. Then the parallel domain $\mathcal{D}(F_1, F_2, F_3)$ is a fundamental domain for the action of the representation in $\mathbb{X}$.

Notice that discreteness and faithfulness is not new, as this follows from classical work of Vinberg [Vin81]. The novelty is the explicit domain in the symmetric space.

The strategy of proof follows the classical strategy used to prove the Poincaré polyhedron theorem, or Vinberg's criterion (see [EP94; Ben09]). One first builds a tesselated space in which W_3 acts with $\mathcal{D}$ as a fundamental domain, and equipped with a local isometry to the symmetric space. Then, one shows that the tesselated space is complete. This step requires the estimates with the parallel domains that we mentioned before.

Orbits of triples of flags in general position under the diagonal action of $SL_3(\mathbb{R})$ are parametrized by the *triple ratio*. The relative character variety sending the peripheral system of the reflection orbifold to regular unipotents $\mathcal{X}^{rel}(W_3, SL_3(\mathbb{R}))$ is thus parametrized by $\mathbb{R} \setminus \{0\}$ via the triple ratio. Positive representations correspond to positive triple ratios.

We then proceed to use these estimates to compute the asymptotics of the critical exponent of ρ_τ associated to a triple of flags with triple ratio τ in the symmetric space. We define the *critical exponent of* ρ in the symmetric space to be

$$h_{\mathbb{X}}(\rho) = \limsup_{T\to\infty} \frac{1}{T} \log \#\{g \in W_3 : d_{\mathbb{X}}(o, \rho(g)o) \leq T\},$$

for some choice of basepoint $o \in \mathbb{X}$. The number is independent of this choice.

Theorem 2 (Theorem 5.1)**.** For each $\tau > 0$, we let ρ_τ be a representation given by a triple of flags with triple ratio τ. Then

$$\lim_{\tau\to+\infty} h_{\mathbb{X}}(\rho_\tau) = \frac{1}{2}.$$

We prove the (stronger) statement that the critical exponent with respect to the Hilbert translation length converges to $\frac{1}{2}$. For $\tau = 1$, the representation preserves an irreducible slice $\mathcal{H}$ with curvature -1, so the critical exponent is one. Theorem

5.1 says that as τ goes to infinity, the critical exponent drops to $1/2$, which coincides with the critical exponent of a cyclic subgroup generated by a regular unipotent element. There are two key ingredients in the proof oxf the theorem above. The first is to use distance estimates in the parallel domain to produce lower bounds for the translation length of elements with little excursion along the cusps compared to their word length. The second ingredient is to obtain a bound for the translation length of those elements with large excursions on the cusps, which we reduce to a computation in the hyperbolic plane (see Section 5.4)

Our computations hint to a positive answer to the following question (see 5.6 and 5.14)

Question: Is the function $\tau \to h_{\mathbb{X}}(\rho_\tau)$ strictly decreasing in $[\tau_0, \infty)$ for some τ_0?

Giuseppe Martone and Harry Bray supervised an undergraduate project [ODN] in which the authors, among other things, performed numerical experiments with the Hilbert critical exponent. In that article, they conjecture the $1/2$ limit at infinity, which the previous result confirms. The numerical data does not settle this question, since critical exponents are hard to estimate for large τ.

Related work. Our motivation came from Schwartz's work on Pappus representations [Sch25], where he associates a pattern of flats with a Pappus representation and builds a sort of pleated surface from it. We arrived at the construction of domains by examining the configurations of flats foliated by medial geodesics that appear in his work. Our ability to fatten these patterns to a hypersurface depends heavily on the positivity of our flag configurations.

There is an extensive list of work providing asymptotics for critical exponents in the Hitchin component, dating back to Nie [Nie15] and Zhang [Zha15]. Recently, in [Car+26] the authors provide asymptotics for specific families of Hitchin representations in terms of Fock-Goncharov parameters on a compact surface with boundary, based on work by Parreau [Par22]. The setup of the previous work is different from ours as they avoid working in the symmetric space, relying on projective techniques instead. In previous work [LL26] we study entropy asymptotics for convex projective structure on pairs of pants, but along sequence with bounded triple ratio, contrary to the case studied here. In upcoming work, we will extend our domains in order to tackle the compact case, by providing a reinterpretation of the Bonahon-Dreyer coordinates in $SL_3(\mathbb{R})$ [BD14].

Recently, the pseudo-Riemannian structure in $\mathbb{Y}$ has been exploited in [WYZ26] in relation to the Axiom A flows constructed in [DMS25].

Organization of the paper. In Section 1 we review classical theory on the symmetric space of $SL_3(\mathbb{R})$. In Section 2, we explore the pseudo-Riemannian structure of the space $\mathbb{Y}$, and introduce the notion of a wall.

Section 3 is the technical heart of the paper. In it, we construct the parallel domains and compute distances between parallel sets in the walls in the boundary, as well as other important distance estimates. Using these results, Section 4 is devoted to proving that the parallel domains are fundamental domains for the representations.

Finally, Section 5 builds the machinery to prove the result on the critical exponent.

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1. The symmetric space of $SL_3(\mathbb{R})$

In this section we will study the geometry of the symmetric space of $SL_3(\mathbb{R})$, which we denote by $\mathbb{X}$. We will only take a brief look at this topic. The reader can find a more thorough treatment in [Don07], [Hel79], and [Ebe96], see also Sambarino's notes on symmetric spaces [Sam24].

1.1. Generalities on the symmetric space.

We define $\mathbb{X}$ as the space of unimodular inner products on $\mathbb{R}^3$, i.e., inner products whose unit ball has the same volume as the standard Euclidean ball.

$SL_3(\mathbb{R})$ acts transitively on $\mathbb{X}$ via $g \cdot q = q \circ g^{-1}$, with stabilizer $SO(q)$ (by definition).

The metric. Denote by $\mathfrak{p}_q$ the space of q-self adjoint, traceless matrices. The assignment

$$X \mapsto \frac{d}{dt}\Big|_{t=0} e^{tX} \cdot q$$

identifies $T_q\mathbb{X} \cong \mathfrak{p}_q$ canonically. Note that the derivative of the action by g is $\mathrm{Ad}(g)$.

We can equip $\mathbb{X}$ with a Riemannian metric by declaring

$$\langle X, Y \rangle_q = \frac{1}{2}\operatorname{tr}(XY),\ \forall X, Y \in \mathfrak{p}_q.$$

Because of the description of the derivative of the action of $SL_3(\mathbb{R})$, we get that $SL_3(\mathbb{R})$ acts by isometries. More specifically, the metric identifies the group with $\mathrm{Isom}_0(\mathbb{X})$. The metric coincides, up to some multiple, with the metric inherited from the Killing form of $SL_3(\mathbb{R})$.

Remark 1.1. The factor of $1/2$ is so that an irreducible totally geodesic copy of $\mathbb{H}^2$ has curvature -1 (see Section 1.4).

One can prove that the geodesic through q tangent to $X \in \mathfrak{p}_q$ is

$$\gamma_{(q,X)}(t) = e^{tX} \cdot q.$$

Geodesics through $q \in \mathbb{X}$ come in two flavors, depending on the velocity vector $X \in \mathfrak{p}_q$. Since X is q-self-adjoint, it is diagonalizable in a q-orthonormal basis. If X has three distinct eigenvalues, we say the geodesic is *regular*. Otherwise, we say that the geodesic is *singular*.

Curvature. If we denote by K the sectional curvature, one can prove (see, for example, [Sam24]) the following formula for the sectional curvature:

$$K(X,Y) = \frac{2\operatorname{tr}([X,Y]^2)}{\operatorname{tr}(X^2)\operatorname{tr}(Y^2) - \operatorname{tr}(XY)^2}. \tag{1}$$

A quick computation shows that this number is always non-positive, and in particular, $\mathbb{X}$ is a $CAT(0)$-space.

1.2. **Flats.** Given $q \in \mathbb{X}$, consider the subspace $\mathfrak{a} \subseteq \mathfrak{p}_q$ of matrices diagonal in a q-orthogonal basis of $\mathbb{R}^3$. It is not hard to see that the image of the map

$$\begin{aligned} \mathfrak{a} &\to \mathbb{X} \\ z &\mapsto e^z \cdot q \end{aligned}$$

is a totally geodesic surface isometric to the euclidean plane $\mathbb{E}^2$ (up to some multiple). We will call such a plane a *flat*.

The action of $SO(q)$ is transitive on the set of flats through q. A flat A is then determined by three points on $\mathbb{RP}^2$ in general position, since such a triple defines a basis that is orthogonal for each inner product in A.

If X is tangent to a regular geodesic, there is only one q-orthogonal basis for which X is diagonal, and thus it is contained in a unique flat. A singular geodesic is contained in a circle worth of flats.

1.3. **Dualities.** Every statement involving points and lines on a projective plane has a dual statement obtained by permuting the roles of points and lines. In fact, one can find bijections $D : \mathbb{RP}^2 \to \mathbb{RP}^2_*$ that preserve incidence relations. This means that two lines $\ell_1, \ell_2 \in \mathbb{RP}^2_*$ intersect at $p \in \mathbb{RP}^2$ if and only if $D^{-1}(\ell_1)$ and $D^{-1}(\ell_2)$ are points contained in the line $D(p)$. These bijections are classically referred to as *dualities* or *polarities*.

If $D : \mathbb{R}^3 \to (\mathbb{R}^3)^*$ is a duality and $v, w \in \mathbb{R}^3$, one can show that the bilinear form $B(v,w) = D(v)(w)$ is symmetric.

We can think of dualities either as maps from $\mathbb{RP}^2$ to $\mathbb{RP}^2_*$ respecting incidences, or as the isomorphisms $D : \mathbb{R}^3 \to (\mathbb{R}^3)^*$ such that the bilinear form $B(v,w) = D(v)(w)$ is symmetric. Notice that in this case, B is always non-degenerate.

Non-degenerate symmetric bilinear forms over $\mathbb{R}$ are classified by their signature (n_+, n_-), where n_+ is the dimension of a maximal positive definite subspace, and n_- the dimension of the maximal negative definite space. We will say that a duality is *elliptic* if the signature is $(3,0)$ or $(0,3)$. Otherwise, we will say that the duality is *hyperbolic*.

Dualities act in the symmetric space $\mathbb{X}$. Given D a duality, define a transformation $T_D : \mathbb{X} \to \mathbb{X}$ to be

$$T_D(q)(v,w) = q^*(D(v), D(w)),$$

where q^* is the inner product induced in $(\mathbb{R}^3)^*$ by q. Furthermore we have the following lemma:

Lemma 1.2. *For a duality D, the transformation T_D is an isometry of $\mathbb{X}$.*

Proof. Fix $q_0 \in \mathbb{X}$, and write $q = q_0(A_q \cdot, \cdot)$ and $B = q_0(M_D \cdot, \cdot)$. Notice that:

$$D(v)(w) = B(v,w) = q_0(M_D v, w) = q(A_q^{-1} M_D v, w),$$

This tells us that:

$$T_D(q)(v,w) = q(A_q^{-1}M_D v, A_q^{-1}M_D w),$$

therefore the matrix associated to the inner product $T_D(q)$ (with respect to q_0) is $M_D A_q^{-1} M_D$ (because both M_D and A_q are q_0-self-adjoint).

Taking q_t a curve of inner products with derivative corresponding to $x \in \mathfrak{p}_{q_0}$, we get that:

$$d_{q_0}T_D(X) = -\mathrm{Ad}(M_D^{-1})(X).$$

Observe that conjugation leaves the trace invariant, which proves the claim. □

Remark 1.3. From the above calculation, it follows that an inner product q is fixed by T_D if and only if the associated matrix M_D with respect to q satisfies $M_D^2 = Id$. Furthermore, notice that M_D is q-self adjoint. Both of these properties imply that M_D is an involution in $O(q)$.

Since a duality D corresponds to the symmetric bilinear form $B(v,w) = D(v)(w)$, we can equivalently think of T_D as being induced by B. From now on, given a non-degenerate symmetric bilinear form B on $\mathbb{R}^3$, we write T_B for the isometry of $\mathbb{X}$ induced by the corresponding duality.

After this discussion, notice that every inner product $q \in \mathbb{X}$ induces a duality. The induced map $\iota_q : \mathbb{X} \to \mathbb{X}$ is simply the symmetric involution that fixes q with $d\iota_q = -Id_{T_q\mathbb{X}}$. That is the behavior of any elliptic duality. Hyperbolic dualities will be studied in the next subsection.

1.4. **Hyperbolic slices of the symmetric space.** Totally geodesic submanifolds of symmetric spaces are themselves symmetric spaces. If the submanifolds are two-dimensional, they always have constant curvature. Moreover, Equation (1) discards the possibility of a totally geodesic submanifold of $\mathbb{X}$ having positive curvature (i.e. spheres), and the zero curvature case is explained by the theory of flats. In this subsection, we study totally geodesic surfaces in $\mathbb{X}$ of constant negative curvature. We will call such surfaces *hyperbolic slices*.

Totally geodesic submanifolds through a point $q \in \mathbb{X}$ correspond to *Lie triple systems*, i.e. subalgebras $\mathfrak{m} \subseteq \mathfrak{p}$ such that

$$[\mathfrak{m}, [\mathfrak{m}, \mathfrak{m}]] \subseteq \mathfrak{m}.$$

Note that in this case the group whose Lie algebra is $\mathfrak{g}_\mathfrak{m} = \mathfrak{m} \oplus [\mathfrak{m}, \mathfrak{m}]$ acts transitively on the submanifold. Because a hyperbolic slice needs to be isometric to a rescaling of $\mathbb{H}^2$, and the Lie algebra of $Isom_0(\mathbb{H}^2)$ is $\mathfrak{sl}_2(\mathbb{R})$, we have that $\mathfrak{g}_\mathfrak{m}$ corresponds to an injective Lie algebra morphism $\mathfrak{sl}_2(\mathbb{R}) \to \mathfrak{sl}_3(\mathbb{R})$. Up to conjugation by the adjoint action of $SL_3(\mathbb{R})$, there are only two such morphisms: one irreducible, and one reducible. Each morphism corresponds to a different class of hyperbolic slice, also called irreducible or reducible respectively.

Irreducible hyperbolic slices will be denoted by $\mathcal{H}$, and reducible slices will be denoted by $\mathbb{H}$. Here is a more hands-on description of these slices.
The irreducible slice $\mathcal{H}$: As mentioned in the previous subsection, non-degenerate symmetric bilinear forms induce isometries of the symmetric space $\mathbb{X}$.

Lemma 1.4. *Given B symmetric and bilinear with signature $(1,2)$, then the induced isometry $T_B : \mathbb{X} \to \mathbb{X}$ pointwise fixes an irreducible hyperbolic slice $\mathcal{H}$, with stabilizer $SO(B) \leq SL_3(\mathbb{R})$.*

Proof. Such bilinear form defines a disk $\mathcal{C}_B := \mathbb{P}(B(v,v) > 0)$. This disk inherits a metric from B inducing the classical Klein model of hyperbolic space.

Every point p in the interior of the conic has an associated symmetric involution $\sigma_p \in SO(B)$. More concretely, this involution is defined as the identity restricted to the subspace defined by p, and as $-Id$ restricted to the subspace $p^{\perp_B}$ (perpendicular taken with respect to B).

Using σ_p one can define $q_p(v,w) = B(\sigma_p v, w)$. An easy computation shows that this is an inner product.

Using Remark 1.3, it is easy to check that $T_B q_p = q_p$ for every $p \in \mathcal{C}_B$. These actually comprise all the fixed point set. □

Remark 1.5. Given $p \in \mathcal{C}_B$, we can explicitly write down the involution σ_p. Given $v \in p$, define:

$$\sigma_p(w) = 2\frac{B(w,v)}{B(v,v)}v - w.$$

This formula gives an explicit expression for q_p as:

$$q_p(w_1,w_2) = \frac{1}{B(v,v)}(2B(w_1,v)B(w_2,v) - B(w_1,w_2)B(v,v)).$$

The reducible slice $\mathbb{H}$: Choose a 2-dimensional subspace $\pi \subset \mathbb{R}^3$ and a vector v transverse to π. The pair $([v],[\pi])$ defines a linear involution $\sigma \in SL_3(\mathbb{R})$ (acting as the identity on $\langle v \rangle$ and as $-Id$ on π). The induced isometry of $\mathbb{X}$ fixes the set

$$Fix(\sigma) = \{q \in \mathbb{X} : q(v,\pi) = 0\}$$

of inner products making the splitting $\langle v \rangle \oplus \pi = \mathbb{R}^3$ orthogonal. The further normalization $q(v,v) = 1$ allows us to identify a totally geodesic surface

$$\mathbb{H} = \{q \in \mathbb{X} : q(v,v) = 1,\ v^{\perp_q} = \pi\} \subset Fix(\sigma),$$

which is the reducible slice. Its stabilizer inside $SL_3(\mathbb{R})$ consists of $g \in SL_3(\mathbb{R})$ such that $gv = \pm v$ and $g(\pi) = \pi$.

Remark 1.6. We mentioned before that these two hyperbolic slices are geometrically different by using representation theory. We can also see it using Equation (1): an irreducible slice $\mathcal{H}$ has constant sectional curvature equal to -1 whereas a reducible slice has constant sectional curvature -4 (which is actually the minimum that K can attain).

The following lemma explains the structure of geodesics that belongs to hyperbolic slices:

Lemma 1.7. *Every geodesic contained in $\mathbb{H}$ or $\mathcal{H}$ is regular.*

Proof. Assume γ is a geodesic such that $\gamma \subseteq \mathbb{H}$ for a reducible slice. Let $q \in \gamma$ and let $X = \dot{\gamma}(0) \in \mathfrak{p}_q$. Then X belongs to the image of a reducible embedding $\iota : \mathfrak{sl}_2(\mathbb{R}) \to \mathfrak{sl}_3(\mathbb{R})$. But this means that X is conjugated to a matrix of the form $\mathrm{diag}(a,0,-a)$, which implies it is regular.

The proof for an irreducible slice is analogous. □

According to the proof of Lemma 1.7, geodesics in hyperbolic slices are not only regular, but also medial.

2. Parallel sets and walls in the symmetric space

In this section we study the space $\mathbb{Y}$ of *parallel sets*, a special family of totally geodesic submanifolds of $\mathbb{X}$. Each parallel set is the fixed point set of an involution of $SL_3(\mathbb{R})$, and this identifies $\mathbb{Y}$ with a conjugacy class of involutions. In particular, $\mathbb{Y}$ is a pseudo-Riemannian symmetric space. Its spacelike geodesics correspond to hypersurfaces of $\mathbb{X}$ saturated by parallel sets, which we call *walls*.

2.1. **Linear involutions and the symmetric space structure.** A *linear involution* of a real vector space V is an element $\sigma \in \mathrm{GL}(V)$ such that $\sigma^2 = \mathrm{Id}$. We denote by $\mathrm{Gr}_k(V)$ the Grassmannian of k-dimensional subspaces of V, and write $\mathbb{P}(V) = \mathrm{Gr}_1(V)$ and $\mathbb{P}(V)^* = \mathrm{Gr}_{\dim V - 1}(V)$.

Lemma 2.1. *Every linear involution σ of V is diagonalizable with eigenvalues ± 1. Denoting by $\sigma^{\pm}$ the corresponding eigenspaces, the map $\tau \mapsto (\tau^+, \tau^-)$ is a $\mathrm{GL}(V)$-equivariant bijection between the conjugacy class $\mathcal{C}(\sigma) = \{g\sigma g^{-1} : g \in \mathrm{GL}(V)\}$ and*

$$\mathbb{Y}_\sigma = \{(V^+, V^-) \in \mathrm{Gr}_{\dim \sigma^+}(V) \times \mathrm{Gr}_{\dim \sigma^-}(V) : V^+ \cap V^- = 0\}.$$

Proof. Every $v \in V$ decomposes as $v = v^+ + v^-$, where $v^{\pm} = \frac{1}{2}(v \pm \sigma v)$ satisfies $\sigma v^{\pm} = \pm v^{\pm}$. This proves the first claim.

For $g \in \mathrm{GL}(V)$, the eigenspaces of $g\sigma g^{-1}$ are $g\sigma^{\pm}$, so the map is equivariant. The map $\tau \to (\tau^+, \tau^-)$ is clearly injective and it is surjective since $\mathrm{GL}(V)$ acts transitively on pairs of complementary subspaces of fixed dimensions. □

From now on $V = \mathbb{R}^3$ and $G = SL_3(\mathbb{R})$, with Lie algebra $\mathfrak{g} = \mathfrak{sl}_3(\mathbb{R})$. If $\sigma \in G$ is an involution then the unit determinant condition forces $\dim \sigma^+ = 1$ and $\dim \sigma^- = 2$. We set

$$\mathbb{Y} := \{(p, \ell) \in \mathbb{RP}^2 \times (\mathbb{RP}^2)^* : p \notin \ell\}.$$

By Lemma 2.1, $\mathbb{Y}$ is in G-equivariant bijection with the set of involutions of G. We denote by $\sigma_{(p,\ell)}$ the involution associated to $(p, \ell) \in \mathbb{Y}$.

Remark 2.2. The involution $\sigma_{(p,\ell)}$ can be written explicitly. Pick $v \in p$ and $\alpha \in (\mathbb{R}^3)^*$ with $\ker \alpha = \ell$ and $\alpha(v) = 2$, then

$$\sigma_{(p,\ell)}(w) = \alpha(w)v - w.$$

The pair (v, α) is unique up to replacing it by $(\lambda v, \lambda^{-1}\alpha)$ for $\lambda \in \mathbb{R} \setminus \{0\}$.

Tangent space and the pseudo-Riemannian structure. Using the splitting $\mathbb{R}^3 = p \oplus \ell$, the tangent space of $\mathbb{Y}$ can be identified with:

$$T_{(p,\ell)}\mathbb{Y} = \mathrm{Hom}(p, \ell) \oplus \mathrm{Hom}(\ell, p).$$

We can think of $T_{(p,\ell)}\mathbb{Y}$ inside $\mathrm{End}(\mathbb{R}^3)$ by using the splitting:

$$\mathrm{End}(\mathbb{R}^3) = (\mathrm{End}(p) \oplus \mathrm{End}(\ell)) \oplus (\mathrm{Hom}(p, \ell) \oplus \mathrm{Hom}(\ell, p)),$$

where the first term in the direct sum is fixed by $Ad(\sigma_{(p,\ell)})$, whereas the second term is anti-fixed by it.

Equipping $\mathfrak{g}$ with a multiple of the Killing form:

$$\langle X, Y \rangle = \frac{1}{2} \mathrm{tr}(XY),$$

for any $X, Y \in \mathfrak{g}$, we inherit a pseudo-Riemannian metric on $\mathbb{Y}$ via the decomposition from before. The choice of normalization will become clear later on.

In particular, given $(S,T) \in T_{(p,\ell)}\mathbb{Y}$, we get:

$$\langle (S,T),(S,T)\rangle = \operatorname{tr}(T \circ S),$$

It is easy to see that this bilinear form has signature $(2,2)$ and the G action is isometric by construction.

Symmetric involutions and geodesics. Notice that $\sigma_{(p,\ell)}$ acts on $\mathbb{Y}$ fixing (p,ℓ). Moreover, the differential of the action at $T_{(p,\ell)}\mathbb{Y}$ is precisely $\mathrm{Ad}(\sigma_{(p,\ell)})$ which, as we saw before, is $-Id$. For that reason, we will call $\sigma_{(p,\ell)}$ the *symmetric involution* of $\mathbb{Y}$ corresponding to (p,ℓ).

Given γ a geodesic in $\mathbb{Y}$, a classical argument shows that $\sigma_{\gamma(t/2)}\sigma_{\gamma(0)}$ is a one parameter subgroup of the form $\exp(tX)$. In particular, $\gamma(t) = \exp(tX)(p,\ell)$, for $X \in T_{(p,\ell)}\mathbb{Y}$. We will now classify geodesics up to the G-action.

Notice that the G-stabilizer $H_{(p,\ell)}$ of a point (p,ℓ) is identified with $GL(\ell)$ thought of as linear transformations preserving the splitting $\mathbb{R}^3 = p \oplus \ell$. Thus, to classify geodesics, it suffices to classify $H_{(p,\ell)}$ orbits of vectors in $T_{(p,\ell)}\mathbb{Y}$.

Let $\{e_i\}_{i=1,2,3}$ be the canonical basis of $\mathbb{R}^3$. Using the homogeneous space structure, it is enough to classify the point $([e_1],[e_1^*]) \in \mathbb{Y}$. Notice that in this case:

$$H_{([e_1],[e_1^*])} = \{\begin{pmatrix} \frac{1}{\det g} & 0 \\ 0 & g \end{pmatrix} : g \in GL_2(\mathbb{R})\} \cong GL_2(\mathbb{R}).$$

Moreover, our choice of basepoint allows us to think of $T_{[e_1],[e_1^*]}\mathbb{Y}$ as $\langle e_2, e_3\rangle \oplus \langle e_2^*, e_3^*\rangle$. The group $GL_2(\mathbb{R})$ acts by sending $g(v,\alpha) = (\det g\, gv, \frac{1}{\det g}\alpha \circ g^{-1})$. A direct linear algebra computation shows the following.

Lemma 2.3. *Let $(v,\alpha) \in T_{([e_1],[e_1^*])}\mathbb{Y}$ be non-zero. Then $\langle (v,\alpha),(v,\alpha)\rangle = \alpha(v)$, and:*

- *If (v,α) is spacelike, then it is in the $\mathrm{GL}_2(\mathbb{R})$-orbit of $\sqrt{\alpha(v)}(e_2,e_2^*)$.*
- *If (v,α) is lightlike, then it is in the $\mathrm{GL}_2(\mathbb{R})$-orbit of exactly one of (e_2,e_3^*), $(e_2,0)$ or $(0,e_3^*)$.*
- *If (v,α) is timelike, then it is in the $\mathrm{GL}_2(\mathbb{R})$-orbit of $\sqrt{-\alpha(v)}(e_2,-e_2^*)$.*

In particular, G acts transitively on unit speed spacelike geodesics, and on unit speed timelike geodesics. We will give a more explicit description of the geodesics later on.

Cross ratios and distances. We will now relate the geometry of $\mathbb{Y}$ to a classical *cross ratio invariant*. Let (p_i,ℓ_i) $i=1,2$ be a pair of points in $\mathbb{Y}$, then the points p_1,p_2, together with the intersections of the line $\overline{p_1p_2}$ with ℓ_1 and ℓ_2 can be associated to a classical *cross ratio* in the projective line defined by p_1 and p_2. This is a projective invariant of the pair (p_i,ℓ_i).

More concretely, suppose $p_i = [v_i]$ and $\ell_i = \ker\alpha_i$, for some vectors and functionals. We will define the cross ratio of the pair to be:

$$CR((p_1,\ell_1),(p_2,\ell_2)) = \frac{\alpha_1(v_2)\alpha_2(v_1)}{\alpha_1(v_1)\alpha_2(v_2)}.$$

This number is independent of the choices of v_i and α_i, and it coincides with the synthetic definition we provided in the first paragraph.

The following relates cross ratios with the pseudo-Riemannian geometry we introduced earlier.

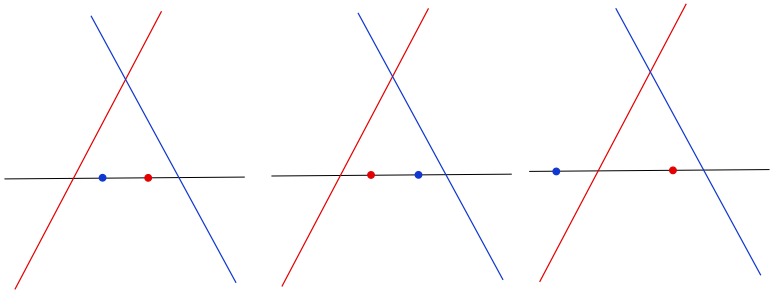


FIGURE 3. The relative positions of a pair in $\mathbb{Y}$. From left to right we call them (ND1), (ND2) and (ND3).

Lemma 2.4. *Let $(p,\ell) \in \mathbb{Y}$, and $X \in T_{(p,\ell)}\mathbb{Y}$ of length ± 1 or zero, then:*

$$CR((p,\ell), \exp(tX)(p,\ell)) = \begin{cases} \cosh^2(t), \text{ if } X \text{ is spacelike,} \\ 1, \text{ if } X \text{ is lightlike,} \\ \cos^2(t), \text{ if } X \text{ is timelike.} \end{cases}$$

Conversely, if $CR((p_1,\ell_1),(p_2,\ell_2)) > 0$, then the points (p_1,ℓ_1) and (p_2,ℓ_2) can be connected by a geodesic in $\mathbb{Y}$.

Proof. The first item follows from an explicit computation with the vectors obtained in Lemma 2.3. The second item follows by noticing that, under a positive cross ratio assumption, we can send the point p_1, p_2, and $\ell_1 \cap \ell_2$ to our favorite canonical basis and verify the claim there. □

As a consequence of the previous lemma, not every pair of parallel sets can be connected by a geodesic in $\mathbb{Y}$. One can, for instance, take a look at pairs (p_i,ℓ_i) in non-degenerate relative position, meaning that p_1, p_2 and the intersections of $\overline{p_1p_2}$ with ℓ_1 and ℓ_2 are distinct.

Figure 3 illustrates three such configurations, labeled (ND1), (ND2), and (ND3), respectively. One can observe that (ND1) and (ND2) verify that $0 < CR < 1$ and $CR > 1$ respectively. On the other hand, (ND3) has a negative cross ratio, thus the pair of points and lines in this configuration cannot be joined by a geodesic.

One can also relate the cross ratio between a pair of points in $\mathbb{Y}$ to the trace of the composition of the associated involutions.

Lemma 2.5. *Let σ_1 and σ_2 be a pair of involutions associated to $(p_1,\ell_1),(p_2,\ell_2) \in \mathbb{Y}$, then:*

$$\operatorname{tr}(\sigma_1\sigma_2) = 4CR((p_1,\ell_1),(p_2,\ell_2)) - 1.$$

Proof. Remark 2.2 allows us to write $\sigma_i = \alpha_i \otimes v_i - Id$ where we normalize $\alpha_i(v_i) = 2$ for $i = 1, 2$. A direct computation shows that $\operatorname{tr}(\sigma_1\sigma_2) = \alpha_1(v_2)\alpha_2(v_1) - 1$. □

Remark 2.6. Notice that $\sigma_1\sigma_2$ always has 1 as an eigenvalue, as it acts trivially in $\ell_1 \cap \ell_2$. Thus the complex eigenvalues of $\sigma_1\sigma_2$ are λ, 1 and λ^{-1} that can be computed using the formula above.

We finish by remarking that there is more to say about the relation of the cross ratio and the geometry of $\mathbb{Y}$. More precisely, Labourie noticed that $\mathbb{Y}$ admits a natural symplectic structure that can be used to recover cross ratios (see [Lab07]). This gives a more conceptual proof of Lemma 2.4.

2.2. **Duality between $\mathbb{X}$ and $\mathbb{Y}$.** We will now construct a correspondence between points in $\mathbb{Y}$ and totally geodesic *parallel sets* on $\mathbb{X}$.

Suppose γ is a singular geodesic in the symmetric space $\mathbb{X}$. This means that $\gamma'(0) \in T_{\gamma(0)}\mathbb{X}$ has a repeated eigenvalue, and thus has an associated pair of transverse eigenspaces $(p_\gamma, \ell_\gamma) \in \mathbb{Y}$ corresponding to the one and two-dimensional eigenspaces, respectively. We will say that γ is *asymptotic* to the pair (p_γ, ℓ_γ).

Unlike a regular geodesic, there is not a unique flat passing through γ. A flat through $\gamma(0)$ is determined by a $\gamma(0)$-orthogonal basis. It contains γ if and only if the basis has the form $\{v_1, v_2, v_3\}$ with $v \in p_\gamma$ and $v_2, v_3 \in \ell_\gamma$ We define the *parallel set* associated to the singular geodesic γ to be:

$$P_\gamma = \bigcup \{F : F \text{ flat}, \gamma \subset F\}.$$

Notice that there is a circle's worth of such flats, as two q-orthogonal bases with two vectors in ℓ_γ are related by an $SO(q|_{\ell_\gamma}) \cong S^1$ transformation.

It is a general fact that parallel sets are always totally geodesic submanifolds (see for instance [Ebe96, Proposition 2.20.12]). We will explain this with a self-contained argument that connects parallel sets to the space $\mathbb{Y}$.

As noted above, we can associate a singular geodesic with a canonical point of $\mathbb{Y}$. This point can, in turn, be associated to an involution $\sigma_{(p_\gamma, \ell_\gamma)}$ acting isometrically on the symmetric space. The fixed point set of this involution is precisely the parallel set P_γ.

Proposition 2.7. *Let γ be a singular geodesic of $\mathbb{X}$ asymptotic to a pair $(p_\gamma, \ell_\gamma) \in \mathbb{Y}$, and let σ_γ be the involution associated with this pair. Then the parallel set P_γ coincides with the fixed point set of σ_γ, and it is a totally geodesic submanifold isometric to a product $\mathbb{H}^2 \times \mathbb{R}$, where $\mathbb{H}^2$ is equipped with the hyperbolic metric of curvature -4. Conversely, any totally geodesic copy of $\mathbb{H}^2 \times \mathbb{R}$ is the parallel set of a singular geodesic.*

Proof. Notice that an involution σ associated to (p, ℓ) fixes an inner product $q \in \mathbb{X}$ if and only if $\sigma \in SO(q)$. This happens precisely when the orthogonal complement of p coincides with ℓ, thus:

$$\mathrm{Fix}(\sigma) = q \in \mathbb{X} : p^{\perp_q} = \ell.$$

Clearly, the whole parallel set P_γ is contained in $\mathrm{Fix}(\sigma_\gamma)$ by construction. Conversely, any inner product fixed by σ_γ restricted to ℓ_γ must share an orthogonal basis with $\gamma(0)|_\ell$.

Notice that unimodular inner products that make the splitting $p_\gamma \oplus \ell_\gamma$ orthogonal are in bijection with (not necessarily unimodular) inner products on ℓ_γ by restriction. The space of inner products on a two-dimensional vector space is isometric to $\mathbb{H}^2 \times \mathbb{R}$, after identifying the space of unimodular inner products with $\mathbb{H}^2$. An explicit computation using Equation 1 shows that the curvature of $\mathbb{H}^2$ is -4.

Finally, notice that any totally geodesic submanifold isometric to $\mathbb{H}^2 \times \mathbb{R}$ verifies that $z_0 \times \mathbb{R}$ is a geodesic contained in infinitely many flats of the form $\gamma \times \mathbb{R}$, for γ a geodesic in $\mathbb{H}^2$ passing through z. Thus, this geodesic is singular. □

Therefore, the space $\mathbb{Y}$ is in bijection with the set of *parallel sets* of $\mathbb{X}$. We will now see that the relative position of pairs of parallel sets reflects on the relative position of the corresponding points of $\mathbb{Y}$ in terms of its geometry.

Proposition 2.8. *Let P_1, P_2 be the parallel sets associated to $(p_1, \ell_1), (p_2, \ell_2) \in \mathbb{Y}$, let σ_1, σ_2 be the corresponding involutions, and suppose that $CR((p_1, \ell_1), (p_2, \ell_2)) \notin \{0, 1\}$. Then $d_{\mathbb{X}}(P_1, P_2)$ is realized by a pair of points, and:*

- *$P_1 \cap P_2 \neq \emptyset$ if and only if $0 < CR < 1$. In this case, (p_1, ℓ_1) and (p_2, ℓ_2) are joined by a timelike geodesic.*
- *If $CR > 1$, then (p_1, ℓ_1) and (p_2, ℓ_2) are joined by a spacelike geodesic, and $CR = \cosh^2 d_{\mathbb{X}}(P_1, P_2)$.*
- *If $CR < 0$, then (p_1, ℓ_1) and (p_2, ℓ_2) are not joined by a geodesic, and $CR = -\sinh^2 d_{\mathbb{X}}(P_1, P_2)$.*

Proof. By Lemma 2.5, the eigenvalues of $\sigma_1\sigma_2$ are $1, \lambda, \lambda^{-1}$ with $\lambda + \lambda^{-1} = 4CR - 2$. Since $CR \notin \{0, 1\}$, these are distinct, so $\sigma_1\sigma_2$ is diagonalizable over $\mathbb{C}$.

If λ is complex, it is necessarily in the unit circle, and thus $\sigma_1\sigma_2$ preserves some inner product $q \in \mathbb{X}$. Otherwise, λ is real, $\sigma_1\sigma_2$ is diagonal over $\mathbb{R}$, and preserves a flat $F_{\sigma_1\sigma_2}$ in the symmetric space. In any case, the translation distance of $\sigma_1\sigma_2$ is $\log|\lambda|$ (with our normalization of the metric on $\mathbb{X}$). Given C the set of points with minimal displacement:

$$C = \{q \in \mathbb{X} : d(q, \sigma_1\sigma_2 q) = \log|\lambda|\},$$

we saw that C is non-empty in both cases, and it is convex, as it is either a flat, or the set of inner products making $\sigma_1\sigma_2$ an isometry.

We claim that $d_{\mathbb{X}}(P_1, P_2) = \log|\lambda|/2$, and that it is attained. For $x_1 \in P_1$ and $x_2 \in P_2$ we have

$$d(x_2, \sigma_1\sigma_2 x_2) = d(x_2, \sigma_1 x_2) \leq d(x_2, x_1) + d(x_1, \sigma_1 x_2) = 2d(x_1, x_2),$$

so $\log|\lambda| \leq 2d_{\mathbb{X}}(P_1, P_2)$. Conversely, σ_1 conjugates $\sigma_1\sigma_2$ with its inverse, so σ_1 preserves C. The midpoint of y and $\sigma_1 y$, for any $y \in C$, is then a point $x_1 \in P_1 \cap C$. Since $d(x_1, \sigma_2 x_1) = d(\sigma_1 x_1, \sigma_1\sigma_2 x_1) = \log|\lambda|$, the midpoint m of $[x_1, \sigma_2 x_1]$ is fixed by σ_2, so $m \in P_2$ and $d_{\mathbb{X}}(P_1, P_2) \leq d(x_1, m) = \log|\lambda|/2$.

Therefore $\log|\lambda| = 2d_{\mathbb{X}}(P_1, P_2)$. If $0 < CR < 1$, then λ lies on the unit circle, and $P_1 \cap P_2 \neq \emptyset$. If $\lambda > 0$, then $4CR - 2 = 2\cosh 2d_{\mathbb{X}}(P_1, P_2)$, that is, $CR = \cosh^2 d_{\mathbb{X}}(P_1, P_2)$. The case where $\lambda < 0$ is identical. The statements about geodesics follow from Lemma 2.4. □

The cases where $CR \in \{0, 1\}$ are somewhat degenerate, and we leave it for the reader to analyze. We note that the $CR = 1$ scenario includes the case of lightlike geodesics, which correspond to asymptotical parallel sets.

One could also go the other way and observe that points in $\mathbb{X}$ define totally geodesic submanifolds of $\mathbb{Y}$, though we will not explore this here. More generally, one can play these games whenever one deals with a *reflective submanifold* of a symmetric space of an arbitrary semisimple Lie group. By reflective we mean that it is the fixed point of an involution.

2.3. **Irreducible slices and their orthogonal parallel sets.**

Our interest in the space $\mathbb{Y}$ arose from studying a special type of totally geodesic submanifold of $\mathbb{X}$ that we call *irreducible hyperbolic slices*. Such slices $\mathcal{H}$ are copies of the hyperbolic plane, and their G-stabilizer is $SO(B)$, for B a non-degenerate symmetric bilinear form of signature $(1, 2)$. The key remark is that the closest point projection $\pi : \mathbb{X} \to \mathcal{H}$ onto a slice $\mathcal{H}$ defines a fibration of $\mathbb{X}$ with parallel sets as fibers.

Let B be a non-degenerate symmetric bilinear form (of arbitrary signature). Such a tensor can be used to define a map of $\mathbb{X}$ via:

$$I_B(q)(v,w) = q^*(B(v,\cdot),B(w,\cdot)) = q([B]_q\cdot,[B]_q\cdot),$$

where q^* is the induced inner product in $(\mathbb{R}^3)^*$, and $[B]_q$ is the element such that $B(\cdot,\cdot) = q([B]_q\cdot,\cdot)$. This is well-defined as long as $\det[B]_q = 1$ for some (thus, for all) $q \in \mathbb{X}$ which we assume from now on. The determinant condition forces B to be an inner product, or have signature $(1,2)$.

Notice that for any $g \in G$, we get that $I_B(gq) = (g^{*_B})^{-1}q$, for $*_B$ the adjoint with respect to B. In particular, we get that I_B is an involution, and $d_qI_B(X) = -X^{*_B}$. This second condition implies that I_B is an isometry. Thus, its fixed point set is totally geodesic.

Notice that when $q \in \mathbb{X}$ is a point in the symmetric space, the map I_q is the *symmetric involution* around q, that is, the map fixing q and acting as $-Id$ in $T_q\mathbb{X}$.

Remark 2.9. Given $q \in \mathbb{X}$ and B a non-degenerate bilinear forms with well-defined I_B, then $I_BI_qI_B = I_{I_B(q)}$. In particular, if $I_B(q) = q$ we get that I_BI_q is again, an involution. Moreover, I_qI_B equals multiplication by $[B]_q$.

When B is not positive definite, we denote the fixed point set of I_B by $\mathcal{H}_B$. More concretely, we can write:

$$\mathcal{H}_B = \{q \in \mathbb{X} : [B]_q \in SO(q)\}.$$

We will call such a submanifold an *irreducible hyperbolic slice*. Being a totally geodesic submanifold, it is necessarily a symmetric space. Moreover, $SO(B)$ acts on the fixed point set $\mathcal{H}_B$, and it acts transitively. In particular, $\mathcal{H}_B$ must be isometric to the symmetric space of $SO(B)$, which is the hyperbolic plane. Applying Equation 1 shows that, with our normalizations, the slice has constant curvature -1.

The following simple proposition motivates most of this paper:

Proposition 2.10. *Let B be a symmetric bilinear form of signature $(1,2)$ with unit determinant defining an ellipse $\mathcal{E}_B = \mathbb{P}(\{v : B(v,v) > 0\})$, and $\pi : \mathbb{X} \to \mathcal{H}_B$ be the closest point projection. For $p \in \mathcal{E}_B$, let σ_p be the involution associated to $(p, p^{\perp_B}) \in \mathbb{Y}$, and $P_p = \mathrm{Fix}(\sigma_p)$ its parallel set. Then:*

1. *The map $p \mapsto q_p = B(\sigma_p\cdot,\cdot)$ is a bijection $\mathcal{E}_B \to \mathcal{H}_B$.*
2. *$T_{q_p}P_p = (T_{q_p}\mathcal{H}_B)^\perp$ and $\pi^{-1}(q_p) = P_p$. In particular, π is a fibration whose fibers are parallel sets.*
3. *The map $\Phi : \mathcal{H}_B \to \mathbb{Y}$, $q_p \mapsto (p, p^{\perp_B})$, is an isometric embedding onto a totally geodesic spacelike surface.*

Proof. Recall that $[B]_q$ is q-self-adjoint, so $q \in \mathcal{H}_B$ if and only if $[B]_q^2 = Id$, that is, $[B]_q$ is a q-orthogonal involution.

1. Let $q \in \mathcal{H}_B$ and (p,ℓ) be the eigenspaces of $[B]_q$. For $v \in p$ and $w \in \ell$ we have

$$B(v,v) = q(v,v) > 0, \qquad B(w,w) = -q(w,w) < 0, \qquad B(v,w) = q(v,w) = 0.$$

Since B has signature $(1,2)$, this forces $\dim p = 1$, $p \in \mathcal{E}_B$ and $\ell = p^{\perp_B}$. Hence $[B]_q = \sigma_p$ and $q = B(\sigma_p\cdot,\cdot) = q_p$. Conversely, the same three identities show that q_p is positive definite. The map is injective because p is the precisely the line fixed by σ_p.

(2) By Remark 2.9, I_B and I_{q_p} commute, and $I_B I_{q_p} = \sigma_p$. The differential $d_{q_p} I_B$ is Id on $T_{q_p}\mathcal{H}_B$ and $-Id$ on its orthogonal complement, while $d_{q_p} I_{q_p} = -Id$. So $d_{q_p}\sigma_p$ is $-Id$ on $T_{q_p}\mathcal{H}_B$ and Id on $(T_{q_p}\mathcal{H}_B)^\perp$, and therefore $T_{q_p}P_p = (T_q\mathcal{H}_B)^\perp$. Notice that $\pi(x) = q$ if and only if the geodesic from q to x is orthogonal to $\mathcal{H}_B$ at q. Hence $\pi^{-1}(q) = P_p$.

(3) The map Φ is $SO(B)$-equivariant, so for $X \in T_q\mathcal{H}_B$ it sends the geodesic $e^{tX}\cdot q$ to $e^{tX}\cdot(p, p^{\perp_B})$. Since X is in the -1 eigenspace of $\mathrm{Ad}(\sigma_p)$, the latter is a geodesic of $\mathbb{Y}$ with initial velocity X. Both metrics are $\frac{1}{2}\operatorname{tr}(X^2)$, which is positive on symmetric matrices, so Φ is a spacelike isometric immersion. Injectivity follows from the first item.

□

In particular, the preimage of a point under the closest point projection $\pi : \mathbb{X} \to \mathcal{H}$ is a parallel set.

There is a more geometric way of thinking of this construction. We can define a hyperboloid $\mathcal{C}_B = \{v \in \mathbb{R}^3 : B(v,v) = 1\}$. The tangent space to $\mathcal{C}_B$ at a point v is precisely the B-orthogonal complement of the line defined by v, which is a negative definite subspace. In particular, $-B|_{T\mathcal{C}_B}$ is an inner product. We can extend it to an inner product in $\mathbb{R}^3$ by extending it as $B|_{\langle v\rangle}$. This extension associated to B is exactly $q_{\langle v\rangle}$.

As in the previous section, one can prove the proposition more generally. More precisely, given a reflective totally geodesic submanifold in a symmetric space, the orthogonal of the tangent space at a point is again a reflective submanifold.

2.4. **Spacelike geodesics revisited.** Proposition 2.10 gives us a synthetic description of spacelike geodesics. More precisely, pulling back geodesics in $\mathcal{H}_B$ to $\mathcal{E}_B$ gives a segment in $\mathcal{E}_B$, corresponding to a geodesic in the Klein model of hyperbolic space. The trace of the corresponding geodesic in $\mathbb{Y}$ corresponds to taking points on this line and their orthogonal complement. Any spacelike geodesic has this form, since two spacelike geodesics lie in the same G-orbit.

We can compactify the space $\mathbb{Y}$ by considering $\mathbb{RP}^2 \times (\mathbb{RP}^2)^*$, and think of the space of full flags $\mathcal{F}$ as its boundary. Given $\gamma : \mathbb{R} \to \mathbb{Y}$, a geodesic obtained from an ellipse B, we see that $\lim_{t\to\pm\infty}\gamma(t) \in \mathcal{F}$ is a pair of flags tangent to the ellipse $\partial\mathcal{E}_B$ at the endpoints of the segment defined by the geodesic. Moreover, these two flags are in general position.

Conversely, we can explicitly compute the geodesics asymptotic to a pair (F_1, F_2). First, we observe that two ellipses tangent to a given pair of flags, and sharing a segment joining the points of the flag define the same spacelike geodesics.

Lemma 2.11. *Let $F_i = (p_i, \ell_i)$, $i = 1, 2$, be transverse flags, and choose $v_i \in p_i$ and α_i with $\ker\alpha_i = \ell_i$ such that $\alpha_1(v_2) = \alpha_2(v_1) = 1$. Let B be a form of signature $(1,2)$ whose conic is tangent to F_1 and F_2, and let $v \in p_1 \oplus p_2$ with $B(v,v) \neq 0$. Then $[v]^{\perp_B} = [\eta_v]$, where*

$$\eta_v = \alpha_1(v)\,\alpha_2 + \alpha_2(v)\,\alpha_1.$$

In particular, $[v]^{\perp_B}$ does not depend on B.

Proof. Since G acts transitively on pairs of transverse flags and the formula for η_v is equivariant, we may assume $F_1 = ([e_1], [e_3^*])$, $F_2 = ([e_3], [e_1^*])$. Tangency means $B(e_1, \cdot)$ and $B(e_3, \cdot)$ are proportional to e_3^* and e_1^* respectively, so up to scaling B

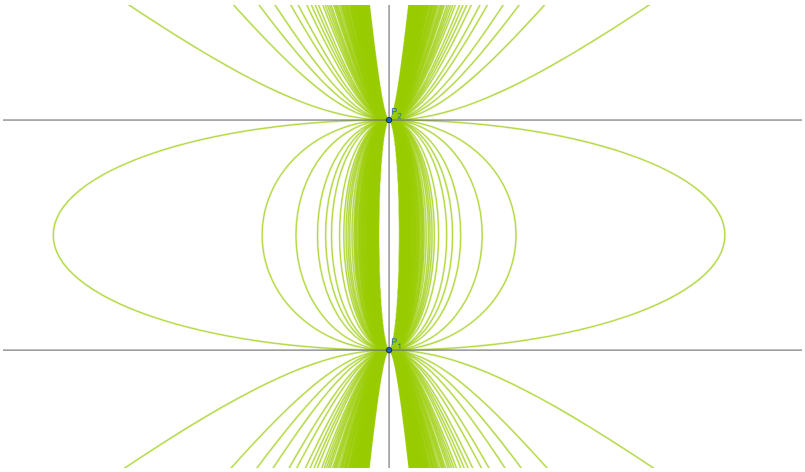


FIGURE 4. Pencil of conics tangent to F_1 and F_2

is an element of the family $B_\lambda(x,y,z) = \lambda xz - y^2$ for some $\lambda \neq 0$ (see Figure 4). For $v = e_3 + se_1$ we get $B_\lambda(v,\cdot) = \frac{\lambda}{2}(e_1^* + se_3^*) = \frac{\lambda}{2}\,\eta_v$. □

The following gives us an explicit expression for a spacelike geodesic asymptotic to a pair of flags in general position.

Proposition 2.12. *Let F_1, F_2 and v_i, α_i be as in Lemma 2.11. Let c and c' be the connected components of $\mathbb{RP}^2 \setminus (\ell_1 \cup \ell_2)$, with $[v_2 + sv_1] \in c$ for $s > 0$. Then:*

(1) *The curves*

$$\gamma_c(t) = ([v_2 + e^{2t}v_1], [\alpha_2 + e^{2t}\alpha_1]),\ \gamma_{c'}(t) = ([v_2 - e^{2t}v_1], [\alpha_2 - e^{2t}\alpha_1]),$$

are unit speed spacelike geodesics of $\mathbb{Y}$, converging to F_2 as $t \to -\infty$ and to F_1 as $t \to +\infty$. They are the only spacelike geodesics of $\mathbb{Y}$ with these limits.

(2) *For every $t, t' \in \mathbb{R}$,*

$$CR(\gamma_c(t), \gamma_c(t')) = \cosh^2(t - t'),\ CR(\gamma_c(t), \gamma_{c'}(t')) = -\sinh^2(t - t').$$

In particular, the parallel sets of $\gamma_c(t)$ and $\gamma_{c'}(t')$ are at distance $|t - t'|$, and they intersect when $t = t'$.

Proof. Writing $v_s = v_2 + sv_1$, Lemma 2.11 gives $\eta_{v_s}(v_{s'}) = s + s'$, and $\eta_{v_s}(v_s) = 2s$, thus:

$$CR((v_s, \eta_{v_s}), (v_{s'}, \eta_{v_{s'}})) = \frac{(s+s')^2}{4ss'}.$$

Substituting $s = \pm e^{2t}$ and $s' = \pm e^{2t'}$ gives the formulas in (2). The limits in (1) are immediate.

Choose B in the pencil of Lemma 2.11 whose ellipsoid $\mathcal{E}_B$ contains the segment $s = \{[v_2 + sv_1] : s > 0\} \subset c$. By Proposition 2.10, $\{(p, p^{\perp_B}) : p \in s\}$ is the image of a geodesic of $\mathcal{H}_B$, hence a spacelike geodesic of $\mathbb{Y}$. The same argument applies for $\gamma_{c'}$. These geodesics are different because when $t = t'$, $\gamma_c(t)$ intersects $\gamma_c(t')$, but are different parallel set. On the other hand, Lemma 2.11 tells us that any spacelike geodesic asymptotic to F_1 and F_2 coincides with γ_c or $\gamma_{c'}$. □

2.5. **The Walls.** We will now use the duality between $\mathbb{X}$ and $\mathbb{Y}$ to define a special type of hypersurface in $\mathbb{X}$ that we call a *wall*. Each parallel set is three-dimensional; thus, a one-parameter family of disjoint parallel sets is automatically codimension one.

As seen in the previous section, any spacelike geodesic in $\mathbb{Y}$ is asymptotic to a pair of transverse flags (F_1, F_2). Conversely, any pair of transverse flags is asymptotic to precisely two spacelike geodesics. A unique geodesic is specified once we fix a favorite connected component c of $\mathbb{RP}^2 \setminus (l_1 \cup l_2)$ containing the points defined by the geodesic in $\mathbb{Y}$.

Definition 2.13. Given a pair of transverse flags (F_1, F_2) and a conected component c as above, we define its associated *wall* as:

$$\mathcal{W}(F_1, F_2, c) = \bigcup_{t \in \mathbb{R}} P(t),$$

where $P(t)$ are parallel sets defined by the spacelike geodesic joining F_1 and F_2 corresponding to the component c.

We sometimes denote $\mathcal{W}(F_1, F_2)$ when c is uniquely determined. By everything we did before, we observe that parallel sets in the definition of $\mathcal{W}(F_1, F_2, c)$ are pairwise disjoint. Since $\mathbb{X}$ is five-dimensional, the walls are hypersurfaces.

The following fact will be useful later on.

Remark 2.14. Any wall $\mathcal{W} \subset \mathbb{X}$ is an analytic hypersurface. In particular, a compact geodesic segment in $\mathbb{X}$ is either contained in $\mathcal{W}$ or intersects it finitely many times.

Any wall can be associated to a one-parameter family of irreducible hyperbolic slices, corresponding to ellipses tangent to F_1 and F_2 and contained in c. Pick any $\mathcal{H}$ in this family, and consider $\pi : \mathbb{X} \to \mathcal{H}$ the closest point projection, then $\mathcal{W}(F_1, F_2, c)$ is the preimage of a geodesic γ asymptotic to F_1 and F_2. Particularly, we get the following.

Lemma 2.15. *Walls $\mathcal{W}(F_1, F_2, c)$ separate the symmetric space into two connected components, each isometric to $\pi^{-1}(H)$, for $\pi : \mathbb{X} \to \mathcal{H}$ the projection onto an irreducible hyperbolic slice, and $H \subset \mathcal{H}$ a half-space.*

Inspired by [Gol99, Chapter 4], we will call the parallel sets $P(t)$ the *slices* of the wall $\mathcal{W}(\gamma, \mathcal{H})$.

The wall construction is natural, in the following sense:

Lemma 2.16. *Suppose $g \in G$ is an element sending a triplet (F_1, F_2, c) as before to (F_1', F_2', c'), then $g\mathcal{W}(F_1, F_2, c) = \mathcal{W}(F_1', F_2', c')$.*

Proof. Any such g sends an ellipsoid tangent to F_1 and F_2 and contained in c to an ellipsoid tangent to F_1' and F_2' and contained in c'. □

Thanks to our understanding of the relative position of parallel sets associated to points in $\mathbb{Y}$, we can develop geometric criteria to verify that walls are disjoint.

Proposition 2.17. *Let $\mathcal{W}(F_1, F_2, c)$ and $\mathcal{W}(F_1', F_2', c')$ be a pair of walls, and let $\sigma \subset c$ and $\sigma' \subset c'$ be the open segments joining the points of the flags corresponding to each wall. Then the walls are disjoint if and only if $\sigma \subset c'$, $\sigma' \subset c$, and $\sigma \cap \sigma' = \emptyset$.*

Proof. □

2.6. **A technical lemma.** For the proof of , we will need the following result explaining how parabolic elements acts on sequences contained in a wall.

Lemma 2.18. *Let $\mathcal{W} = \mathcal{W}(F, F')$ be a wall, and let h be a parabolic fixing F. Assume that $(p_n, \ell_n) \in \mathcal{N}$ are such that the parallel sets $P(p_{n_k}, \ell_{n_k})$ are included in $\mathcal{W}$ and converge to F in $\mathbb{RP}^2 \times (\mathbb{RP}^2)^*$. Then $h^k(p_{n_k}, \ell_{n_k})$ converge to F.*

Proof. Note that the action of $SL_3(\mathbb{R})$ is transitive on pairs of flags in general position, thus, we can assume that $F = F_3$ and $F' = F_1$.

Recall that $P(p, \ell) \subseteq \mathcal{W}$ is equivalent to asking that p is in the line joining P_1 with P_3, and that $\ell = p^\perp$, with respect to any bilinear $(2,1)$ form tangent to F_1, F_3.

We fix the triple $F_3 = ((1, s, 0), e_3)$. Note that h is conjugated to

$$\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix},$$

thus, up to a change of basis, we may assume

$$h = \begin{pmatrix} 2 & -1/s & 1 \\ s & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Let p_u be a point in the segment joining F_1 and F_3, then $p_u = (1, s+u, us)$ with u varying between 0 and $+\infty$. We will prove the stronger result that, when $u \to 0$ and $n \to \infty$, we have that $h^n(p_u, p_u^\perp) \to F_3$.

A quick calculation shows

$$h^n = \begin{pmatrix} n+1 & -n/s & \frac{n^2+n}{2} \\ ns & -n+1 & \frac{n^2 s - ns}{2} \\ 0 & 0 & 1 \end{pmatrix}, \qquad h^n p = \begin{pmatrix} \frac{-2nu + n^2 s^2 u + n s^2 u}{2s} \\ \frac{-2nu + n^2 s^2 u - n s^2 u}{2} \\ su \end{pmatrix}.$$

By a careful analysis of the three cases: when $n^2 u \to 0$, when $nu \to 0$ and $n^2 u$ does not converge to 0, and when $n^2 u \to \infty$, we get that the limit is always $(1, s, 0)$.

Because of Lemma 2.11, we get that $p^\perp = e_3^* + use_1^*$. We identify this functional with the vector $q_u = (us, 0, 1)$, and thus we get that $h^n \cdot p^\perp$ corresponds to $(h^{-n})^t q_u = (-nsu + su, nu, \frac{n^2 su - nsu + 2}{2})$.

Again analyzing the three regimes we get that $h^n \cdot p^\perp$ converges to e_3^*, which proves the lemma. □

3. Parallel domains

This section constructs a domain $\mathcal{D}(F_1, F_2, F_3)$ on the symmetric space $\mathbb{X}$ associated with a positive triple of flags (F_1, F_2, F_3) with boundary components consisting of walls. These domains mimic how we associate to a triple of points in $\partial\mathbb{H}^2$ a unique ideal triangle in hyperbolic space.

3.1. **Projective invariants.** First, we will define projective invariants, which will allow us to parametrize the character variety, and then we will construct equivariant objects associated to the representations using the contents of Section 2.5.

Given a triple of flags $F_i = (p_i, \ell_i)$, $i = 1, 2, 3$ flags in general position, take $(v_i, \alpha_i) \in \mathbb{R}^3 \times (\mathbb{R}^3)^*$ projecting to $(p_i, \ell_i) \in \mathcal{F}(\mathbb{R}^3)$ and define the triple ratio as follows:

$$TR(F_1, F_2, F_3) = \frac{\alpha_1(v_2)\alpha_2(v_3)\alpha_3(v_1)}{\alpha_1(v_3)\alpha_2(v_1)\alpha_3(v_2)}.$$

We will say that a triple of flags is *positive* (resp. negative) if they have a positive (resp. negative) triple ratio. Note that a triple of flags is positive if and only if the three flags bound a convex triangle in some affine chart of $\mathbb{RP}^2$ (the lines bound a triangle with the points of the flags in the side).

There is another important invariant called *the cross ratio*. This time, we let $(p_1, \ell_1), (p_2, \ell_2) \in \mathbb{Y}$. We then define, picking any representatives $v_i \in p_i \setminus 0$ and α_i with $\ker \alpha_i = \ell_i$,

$$CR\left((p_1, \ell_1), (p_2, \ell_2)\right) = \frac{\alpha_1(v_2), \alpha_2(v_1)}{\alpha_1(v_1), \alpha_2(v_2)}. \tag{2}$$

3.2. **Triples of flags and its symmetries.** Note that, given two triples F_1, F_2, F_3 and F_1', F_2', F_3', there exists an element $g \in SL_3(\mathbb{R})$ such that $gF_i = F_i'$ if and only if $TR(F_1, F_2, F_3) = TR(F_1', F_2', F_3')$.

From the formula, it is clear that the triple ratio is invariant under cyclic permutation of the flags, i.e., $TR(F_1, F_2, F_3) = TR(F_2, F_3, F_1)$. Therefore, for any triple of flags in general position (F_1, F_2, F_3), there is an $S \in SL_3(\mathbb{R})$ sending $SF_i = F_{i+1}$ (indices mod three).

Let $s \neq 0$ and consider the triple of flags given by

$$\begin{aligned} F_1(s) &= (se_2 + e_3, e_1^*) \\ F_2(s) &= (e_1 + se_3, e_2^*), \\ F_3(s) &= (se_1 + e_2, e_3^*). \end{aligned}$$

where (e_1, e_2, e_3) is the canonical basis.

A quick computation shows that $TR(F_1(s), F_2(s), F_3(s)) = s^{-3}$, thus for every triple of flags there exists a unique s such that $(F_1(s), F_2(s), F_3(s))$ is in its $SL_3(\mathbb{R})$. Under this normalization, the symmetry S is thus defined by $Se_i = e_{i+1}$.

More symmetries can be obtained by considering dualities (see Section 1.3). Notice that, for $i \neq j \neq k$, there is no element of $SL_3(\mathbb{R})$ that fixes F_i and swaps F_j and F_k. This is because

$$TR(F_1, F_2, F_3) = TR(F_1, F_3, F_2)^{-1}.$$

Note that a duality D acts in the space of flags by sending $F = (p, \ell)$ to $DF = (D\ell, Dp)$. A quick computation shows that

$$TR(F_1, F_2, F_3) = TR(DF_1, DF_2, DF_3)^{-1}.$$

Lemma 3.1. *Let F_1, F_2, F_3 be a triple of flags in general position. There is a unique hyperbolic duality that fixes F_3 and permutes F_1 and F_2.*

Proof. Again, we assume that the flags are $F_1(s), F_2(s)$ and $F_3(s)$. Dualities are defined by non-degenerate symmetric bilinear forms B. Write $B(v, w) = q_0(Mv, w)$, for q_0 the standard inner product. We want the following properties:

- $se_2 + e_3$ is $B-$orthogonal to $e_1 \oplus e_3$.
- $e_1 + se_3$ is $B-$orthogonal to $e_2 \oplus e_3$.
- e_3 is $B-$orthogonal to $\langle se_2 + e_3 \rangle \oplus \langle e_1 + se_3 \rangle$.
- The $B-$orthogonal complement of $se_1 + e_2$ is precisely $e_1 \oplus e_2$.

The first three conditions imply that $M(se_2 + e_3) = \lambda e_2$, $M(e_1 + se_3) = \mu e_1$ and $Me_3 = \nu(se_2 + e_3) \times (e_1 + se_3)$, for some $\lambda, \mu, \nu \in \mathbb{R}$. Solving for Me_i we get that:

$$M = \begin{pmatrix} \mu - s^3\nu & -\nu s & \nu s^2 \\ -s\nu & \frac{\lambda-\nu}{s} & \nu \\ s^2\nu & \nu & -\nu s \end{pmatrix}.$$

The last condition implies that $M(se_1+e_2)$ is collinear to e_3. A quick computation shows that this condition implies that $\mu = \lambda = (1 + s^3)\nu$, and thus

$$M = \nu \begin{pmatrix} 1 & -s & s^2 \\ -s & s^2 & 1 \\ s^2 & 1 & -s \end{pmatrix}.$$

Normalizing so that $det(M) = 1$, we get our desired symmetric bilinear form. □

3.3. **Walls and symmetries of representations.** We fix a triple of flags F_1, F_2, F_3 in general position. Let $\mathcal{W}_i = \mathcal{W}(F_i, F_{i-1})$, where we fix the walls with respect to the line L fixes by the symmetry S defined by $SF_i = F_{i+1}$. If $TR(F_1, F_2, F_3) > 0$, then the three flags bound a triangle in an affine chart of $\mathbb{RP}^2$. This choice of walls corresponds to the segments lying inside the triangle.

Remark 3.2. Let $\sigma_i = \sigma_i(F_{i+1}, F_{i-1})$. Note that the segment defining $\mathcal{W}_i$ is invariant under σ_i, and so are the flags, thus, $\sigma_i\mathcal{W}_i = \mathcal{W}_i$.

Recall from Lemma 3.1 that there is a duality D_i that fixes F_i and exchanges F_{i-1} and F_{i+1}.

Lemma 3.3. *Let F_1, F_2, F_3 be a triple of flags in general positions such that $TR(F_1, F_2, F_3) > 0$. Then $D_i\mathcal{W}_i = \mathcal{W}_i$. Moreover, $D_i\mathcal{W}_{i+1} = \mathcal{W}_{i-1}$.*

Proof. We assume $i = 1$ for clarity. Because of Remark **??**, the dual of a wall is still a wall, and because $DF_2 = F_3$, we get that $D\mathcal{W}_1$ is either $\mathcal{W}_1$ or $\overline{\mathcal{W}_1}$, in order to show the former, it suffices to see that the dual of a point p does not intersect the segment defining $\mathcal{W}_1$. Let γ_1 be the segment of $P_2 \oplus P_3$ defining $\mathcal{W}_i$.

Define $P = (P_1 \oplus (L_2 \cap L_3) \cap \gamma)$. Convexity of the triangle defined by the triple of flags implies that P is well-defined (the segment and the line intersect). Note that DP is a line that goes through $L_2 \cap L_3$ and through $Q = (P_2 \oplus P_3) \cap L_1$, note that, because of the convexity of the triangle, Q is outside γ and thus $D\mathcal{W}_1 = \mathcal{W}_1$.

A similar argument, tracking the dual of the point intersecting γ_2 joining $L_1 \cap L_3$ with P_2, shows that $D\mathcal{W}_2 = \mathcal{W}_3$. □

3.4. **Definition of the domains.** Notice that a triple of flags in general position (F_1, F_2, F_3) is positive if and only if the points corresponding to the flags bound a triangle whose interior T is entirely contained in a connected component of the complement of the lines of F_i . This lets us choose, for each i, a canonical connected component c_i of the complement of the lines of F_{i-1}, F_{i+1} containing T. Therefore, for a positive triple, we can assign three walls $\mathcal{W}_i = \mathcal{W}(F_{i-1}, F_{i+1}, c_i)$.

Proposition 2.17 implies that the walls $\mathcal{W}_i$ we just constructed are pairwise disjoint. The following lemma ensures that these three walls bound a unique region.

Lemma 3.4. *Given $\mathcal{W}_i$, $i = 1, 2, 3$ associated with a positive triple (F_1, F_2, F_3) as above, then $\mathbb{X} \setminus \bigcup_{i=1}^{3} \mathcal{W}_i$ is a union of four connected components $\mathcal{D} \cup \Delta_1 \cup \Delta_2 \cup \Delta_3$, where*

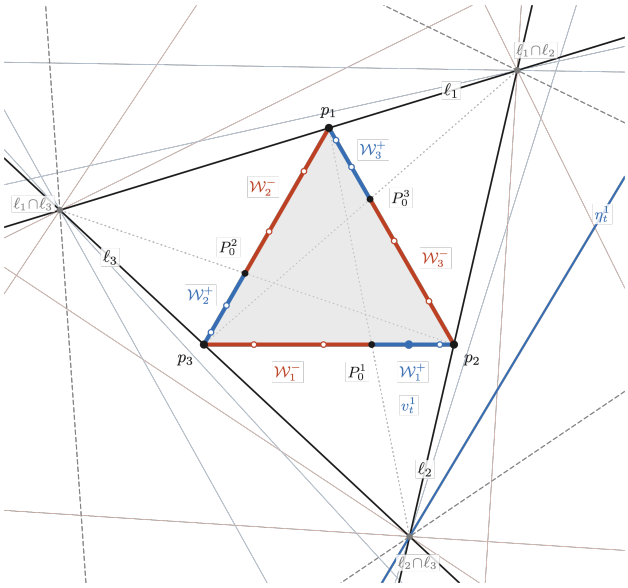


FIGURE 5. Parametrization of the walls $\mathcal{W}^i$.

- *$\partial\mathcal{D} = \bigcup_{i=1}^3 \mathcal{W}_i$ and $\mathcal{D}$ is invariant under the order three involution S that cyclically permutes F_i.*
- *$\partial\Delta_i = \mathcal{W}_i$, and Δ_i is isometric to the preimage of a half-plane in an irreducible copy of $\mathbb{H}^2$ under the closest-point projection.*

Proof. By Lemma **??**, the complement of $\mathcal{W}_i$ has two connected components, each isometric to the preimage of an irreducible half-plane via the closest-point projection. Since the walls are disjoint, $\mathbb{X} \setminus \bigcup_{i=1}^3 \mathcal{W}_i$ has at most four components. The existence of the order three symmetry S permuting the walls forces the existence of a component $\mathcal{D}$ bounded by the three walls simultaneously. □

Given a positive triple of flags (F_1, F_2, F_3), we denote the associated $\mathcal{D}$ of the lemma by $\mathcal{D}(F_1, F_2, F_3)$ and we call it *parallel domain*.

Notice that, by construction, the parallel set P_S invariant by the order three symmetry cyclically permuting the flags is contained in $\mathcal{D}(F_1, F_2, F_3)$.

3.5. **Distance estimates.** Parallel domains allow for explicit computations of distances between parallel sets contained in the walls. The formulas developed here will be important in the proof of Theorems this paper.

Let $F_i = (p_i, \ell_i)$, $i = 1, 2, 3$, be an ordered positive triple of flags, defining a triple of walls $\mathcal{W}_i$ defining the parallel domain $\mathcal{D}(F_1, F_2, F_3)$. For each $i \in \{1, 2, 3\}$, define the *base parallel set* P_0^i of $\mathcal{W}_i$ as the parallel set of the spacelike geodesic defining $\mathcal{W}_i$ whose point corresponds to $(p_{i-1} \oplus p_{i+1}) \cap (p_i \oplus (\ell_{i-1} \cap \ell_{i+1}))$ (see Figure 5).

This base parallel set splits $\mathcal{W}_i$ into two connected components, corresponding to two segments of the geodesic in the pseudo-Riemannian space. We will denote the segment converging to F_{i-1} as $\mathcal{W}_i^-$ and to the one converging to F_{i+1} as $\mathcal{W}_i^+$. Using this decomposition, we will parametrize each parallel set of $\mathcal{W}_i$ by declaring that $P \in \mathcal{W}_i^\pm$ gets mapped to $\pm t$, for $CR(P_0^i, P) = \cosh^2(t)$. By Lemma **??**, this

coincides with the signed distance between P and P_0^i along the geodesic of the pseudo-Riemannian symmetric space (which is the distance between the parallel sets in the symmetric space).

Proposition 3.5. *Let $\mathcal{W}_i$, $i = 1, 2, 3$ be the walls corresponding to the positive triple of flags with triple ratio $\tau \neq 0$. Then*

$$CR(P_t^i, P_{t'}^{i+1}) = \frac{\left(1 + e^{2t} + \tau e^{2t'+2t}\right)\left(1 + \tau e^{2t} + \tau e^{2t'+2t}\right)}{4\tau e^{2t'+2t}}.$$

Particularly, since

$$CR(P_t^i, P_{t'}^{i+1}) = \cosh^2\left(d_{\mathbb{X}}(P_t^i, P_{t'}^{i+1})\right),$$

we get an expression for the distance between the parallel sets at the boundary of our domain. Before proving Proposition 3.5, we will see some consequences of this computation.

Corollary 3.6. *Let (F_1, F_2, F_3) be a positive triple of flags inducing the walls $\mathcal{W}_i$ as before, we get that*

$$\inf_{t' \in \mathbb{R}} \cosh\left(d_{\mathbb{X}}(P_t^i, P_{t'}^{i+1})\right) = \frac{\sqrt{1 + e^{2t}} + \sqrt{1 + \tau e^{2t}}}{2}, \tag{3}$$

and the equality is attained at $t' = \frac{1}{4}\log\left(\frac{(1+e^{2t})(1+\tau e^{2t})}{\tau^2}\right) - t$. Moreover, we get the following useful facts:

1. *$d_{\mathbb{X}}(P_{t_n}^i, P_{t'_n}^{i+1}) \to 0$ if and only if $t_n \to -\infty$, $t'_n \to \infty$ and $t_n + t'_n \to -\frac{1}{2}\log\tau$.*
2. *The distance from the half-wall $\mathcal{W}_i^+$ to $\mathcal{W}_{i+1}$ is given by the expression*

$$\cosh\left(d_{\mathbb{X}}(\mathcal{W}_i^+, \mathcal{W}_{i+1})\right) \geq \frac{(\sqrt{2} + \sqrt{1+\tau})}{2} > 1.$$

3. *Given $\tau_0 = \frac{\sqrt{3}-1}{2}$, the distance between the half walls $\mathcal{W}_i^-$,$\mathcal{W}_{i+1}^-$ is given by*

$$\cosh\left(d_{\mathbb{X}}(\mathcal{W}_i^-, \mathcal{W}_{i+1}^-)\right) = \begin{cases} \sqrt{\frac{(2+\tau)(1+2\tau)}{4\tau}}, & \tau \leq \tau_0, \text{ when } (t,t') = (0,0), \\ \frac{\sqrt{2}+\sqrt{1+\tau^{-1}}}{2}, & \tau \geq \tau_0, \text{ when } \left(-\frac{1}{4}\log(2\tau(1+\tau)),\, 0\right), \end{cases}$$

Proof. Write $X = 1 + e^{2t}$, $Y = 1 + \tau e^{2t}$ and $Z = \tau e^{2t'+2t}$, then we can rewrite

$$CR(P_t^i, P_{t'}^{i+1}) = \frac{(X+Z)(Y+Z)}{4Z},$$

For fixed t, the function $f(Z) = (X/Z + 1)(Y + Z) = X + Y + XY/Z + Z$ is strictly convex on $Z > 0$, with minimum at $Z_m = \sqrt{XY}$. Plugging back this minimizer in terms of t' and t, we obtain Equation 3.

The first two items in the list above follow directly from the formula we just derived. Let us prove (3). On the quadrant $\{(t, t') : t \leq 0,\ t' \leq 0\}$, the function $CR(P_t^i, P_{t'}^{i+1})$ is proper, as $Z \to 0$ forces the cross-ratio to go to infinity. Thus, an infimum is attained. This infimum cannot be attained in the interior of the quadrant, as if this were the case in (t_0, t'_0) for $t_0, t'_0 < 0$, then this pair is realizing

$$\inf_{t' \in \mathbb{R}} \cosh\left(d_{\mathbb{X}}(P_{t_0}^i, P_{t'}^{i+1})\right) = \frac{\sqrt{1 + e^{2t_0}} + \sqrt{1 + \tau e^{2t_0}}}{2}.$$

Since it is a global minimum, $\partial_t CR(P_t^i, P_{t'}^{i+1})$ vanishes at (t_0, t'_0), but by the expression above, this is not the case. This remark shows that the infimum is attained

either at $t' = 0$ or $t = 0$. Solving the minimization problem gives the desired result. □

Remark 3.7. As shown in Lemma 3.3, the duality of $\mathcal{F}$ that fixes the flag F_i and swaps F_{i-1} with F_{i+1} preserves $\mathcal{W}_i$ and exchanges $\mathcal{W}_{i-1}$ with $\mathcal{W}_{i+1}$. One can verify that $\mathcal{W}_i^+$ is mapped to $\mathcal{W}_i^-$, and since the triple ratio changes from τ to τ^{-1}, the estimates above show that:

$$\cosh\left(\frac{d_{\mathbb{X}}(\mathcal{W}_i^-, \mathcal{W}_{i-1})}{2}\right) \geq \frac{1}{2}(\sqrt{2} + \sqrt{1+\tau^{-1}}).$$

More generally, all the estimates hold between indices i and $i-1$ by swapping τ by τ^{-1} in every formula.

From Lemma 2.11, we obtain the following calculation: let $F_i = ([v_i], [\alpha_i])$, $i = 1, 2$, be a pair of flags, such that $\alpha_i(v_{i+1}) = 1$, then given $v_t = v_1 + e^t v_2$, we get that if η_v is the perpendicular of v_t $\eta_{v_t} = \alpha_1 + e^t \alpha_2$, and

$$CR((v_t, \eta_{v_t}), (v_{t'}, \eta_{v_{t'}})) = \cosh^2(t - t').$$

Proof of Proposition 3.5. Because $SL_3(\mathbb{R})$ acts by isometry on the symmetric space, it is enough to prove the proposition for each choice of triple ratio of flags.

Let $(e_i)_{i=1,2,3}$ be the canonical basis of $\mathbb{R}^3$, we will prove the proposition for $F_i = ([e_i], [\alpha_i])$ for $\alpha_i = e_{i+1}^* + s e_{i-1}^*$. Note that we this is invariant under the symmetry S defined via $Se_i = e_{i+1}$. Moreover, we have $\tau^{-1/3} = s$, where $\tau = TR(F_1, F_2, F_3)$.

In these coordinates, the point $\overline{p_{i-1}p_{i+1}} \cap (p_i \oplus (\ell_{i-1} \cap \ell_{i+1}))$ is defined by the vector $v_0^i = e_{i+1} + s^2 e_{i-1}$. Thus, we get that

$$v_t^i = e_{i+1} + e^{-2t} s^2 e_{i-1},\ \eta_t^i = \alpha_{i+1} + s e^{-2t} \alpha_{i-1}$$

parametrize the parallel sets P_t^i.

A computation shows that

$$\eta_t^i(v_{t'}^{i+1}) = 1 + s^3 e^{-2t'} + s^3 e^{-2t'-2t},\ \eta_{t'}^{i+1}(v_t^i) = s\left(1 + e^{-2t'} + s^3 e^{-2t'-2t}\right),$$

while $\eta_t^i(v_t^i) = 2s^2 e^{-2t}$ and $\eta_{t'}^{i+1}(v_{t'}^{i+1}) = 2s^2 e^{-2t'}$. Using that $s^3 = \tau^{-1}$ and the definition of the cross ratio, we get the desired expression. □

We can also explicitely compute the distance between P_S and the walls.

Lemma 3.8. *Let P_t a parallel set in the wall $\mathcal{W}_i$, then*

$$CR(P_S, P_t) = \frac{2}{3}\cosh(\log \tau^{1/6})\left(\cosh(\log \tau^{1/6}) + \cosh(2t + \frac{1}{2}\log \tau)\right).$$

Particularly, $d_{\mathbb{X}}(P_S, P_t)$ is minimized when $t = -\frac{1}{4}\log \tau$.

Proof. This follows from a direct computation in the coordinates used to prove Proposition 3.5, and from noticing that P_S is associated with the vector $(1, 1, 1)$ and its orthogonal functional (with respect to the standard inner product). □

Notice that when $\tau = 1$, this recovers that the inradius of an ideal triangle in $\mathbb{H}^2$ is $\cosh^{-1}(2/\sqrt{3}) = \frac{1}{2}\log 3$ by specializing to $\tau = 1$ and $t = 0$. We will need the following remark about the formula given by Proposition 3.5:

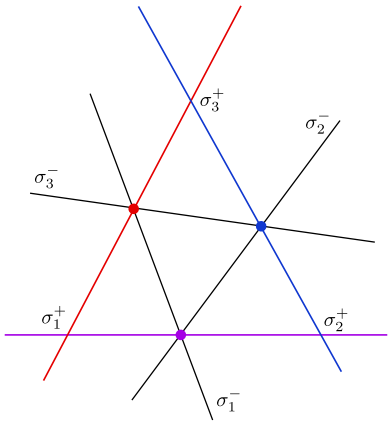


FIGURE 6. How to construct representations of W_3 from a flag.

4. Representations of W_3 and fundamental domains

The goal of this section is to explain how representations of W_3 into $SL_3(\mathbb{R})$ whose length-two elements are parabolic are in bijection with triples on flags, and show that the parallel domains of such triple of flags form a fundamental domain for the action of the group in the symmetric space.

4.1. W_3 **and its representations in** $SL_3(\mathbb{R})$**.** Recall that W_3 is the group generated by reflections along an ideal triangle in $\mathbb{H}^2$. A presentation for this group is

$$W_3 = \langle s_1, s_2, s_3 | s_i^2 \rangle$$

We will be interested in studying representations of W_3 into $SL_3(\mathbb{R})$. Choosing a representation of this group is equivalent to choosing three involutions.

Starting from a triple of flags in general position $F_i = (P_i, L_i)$, $i = 1, 2, 3$, one can construct a representation of the ideal triangle reflection group W_3 into $SL_3(\mathbb{R})$.

The construction goes as follows: define $p_i = L_j \cap L_k$, for $i \neq j \neq k$, and $\ell_i = P_j \oplus P_k$, and consider pairs $(p_i, \ell_i) \in \mathcal{N}$. These pairs define three involutions σ_i via the correspondence described in Subsection 2.1. With these three involutions in hand, let $\rho_{(F_1,F_2,F_3)} : W_3 \to SL_3(\mathbb{R})$ be the representation $\rho_{(F_1,F_2,F_3)}(s_i) = \sigma_i$ (see Figure 6).

Because a pair of involution fixes a flag, this way of constructing representations implies that the product of two involutions will be conjugated to the matrix

$$u = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

With this in mind we define the space of *representations relative to* u, denoted by $Hom_{rel}(W_3, SL_3(\mathbb{R}))$, as the set of representations $\rho \in Hom(W_3, SL_3(\mathbb{R}))$ for which $\rho(s_i s_j)$ is conjugate to u. One can check that $\rho_{(F_1,F_2,F_3)} \in Hom_{rel}(W_3, SL_3(\mathbb{R}))$.

It is not hard to recover the three flags from a representation of W_3, with this fact in mind, we have the following lemma:

Lemma 4.1. *There is a one-to-one correspondence between triples of flags in general position and irreducible representations in* $Hom_{rel}(W_3, SL_3(\mathbb{R}))$.

If $g \in SL_3(\mathbb{R})$, and (F_1, F_2, F_3) is a triple of flags in general position, then $g \circ \rho_{(F_1,F_2,F_3)} \circ g^{-1} = \rho_{(gF_1,gF_2,gF_3)}$. As mentioned in Section 3.1, the triple ratio parametrizes $SL_3(\mathbb{R})$ orbits of triples of flags in general position. Putting everything together, we get that the *relative character variety* $\mathcal{X}_{rel}(W_3, SL_3(\mathbb{R})) = Hom_{rel}(W_3, SL_3(\mathbb{R}))//SL_3(\mathbb{R})$ is isomorphic to $\mathbb{R} \setminus \{0\}$.

We will call the connected component of $\mathcal{X}_{rel}(W_3, SL_3(\mathbb{R}))$ corresponding to positive triples of flags the *relative Hitchin component*. This is the connected component that contains the geometric representation of W_3 into $SO(1,2)$ (See Section 1.3 of [Ben09, Lecture 1]).

Recall that in the previous section we worked with the following parametrization of the orbit of triples of flags, given by, for $s \neq 0$,

$$F_1(s) = (se_2 + e_3, e_1^*)$$
$$F_2(s) = (e_1 + se_3, e_2^*).$$
$$F_3(s) = (se_1 + e_2, e_3^*).$$

We will use the same parametrization for the rest of the paper. This triple defines a representation $\rho_s \in Hom_{rel}(W_3, SL_3(\mathbb{R}))$. Note that every representation in $Hom_{rel}(W_3, SL_3(\mathbb{R}))$ is conjugated to a unique ρ_s for some $s \in \mathbb{R} \setminus \{0\}$.

Recall also that given $\rho \in Hom_{rel}(W_3, SL_3(\mathbb{R}))$, we can find an $S \in SL_3(\mathbb{R})$ such that $S\rho(s_i)S^{-1} = \rho(s_{i+1})$. In the normalization above, $S \in SL_3(\mathbb{R})$ is simply the transformation defined by $Se_i = e_{i+1}$.

Now we are ready to state the main theorem of the paper, whose proof will be developed in the rest of the section.

Theorem 4.2. *Let (F_1, F_2, F_3) be a positive triple of flags and $\rho : W_3 \to SL_3(\mathbb{R})$ be the representation associated to this triple, then the parallel domain $\mathcal{D}$ defined by Lemma 3.4 associated to the triple is a fundamental domain for the action of ρ in the symmetric space $\mathbb{X}$.*

4.2. **The tessellated space and the developing map.** Let (F_1, F_2, F_3) be a positive triple of flags defining a representation ρ, and with domain $\mathcal{D}$ as in the previous section.

The proof of Theorem 4.2 follows the usual strategy used to prove the Poincaré polyhedron theorem. This is the same proof used in generalizations like Vinberg's Theorem, see, for example, [Ben09]. A crucial element of this proof is the introduction of the *tessellated space*.

$$\text{Tess}(\rho) := W_3 \times \overline{\mathcal{D}}/\sim,$$

where we define the equivalence relation $(g, x) \sim (gs_i, \rho(s_i)x)$ for $x \in \mathcal{W}_i$. This space comes equipped with a natural *developing map* $\text{Dev} : \text{Tess}(\rho) \to \mathbb{X}$ defined by sending the equivalence class of $[(g, x)] \to \rho(g)x$. The equivalence relation makes this map well-defined.

The space $\text{Tess}(\rho)$ is a manifold, and inherits the riemannian metric from the symmetric space. To see this, notice that if $x \in \mathcal{D}$ is an interior point, then any equivalence class $[(g, x)]$ has a unique representative, and therefore Dev maps it bijectively to a neighborhood of the symmetric space. If $x \in \mathcal{W}_i$, then choosing U a $\rho(s_i)$-invariant neighborhood disjoint from $\mathcal{W}_{i-1} \cup \mathcal{W}_{i+1}$ (which exists because $\rho(s_i)$ is an involution), allows us to define $\mathcal{U} \subset \text{Tess}(\rho)$ as

$$\mathcal{U} = \{[(g, x)] : x \in U \cap \overline{\mathcal{D}}\} \cup \{[(gs_i, x)] : x \in U \cap \overline{\mathcal{D}}\}.$$

One can verify that $\mathrm{Dev}|_{\mathcal{U}} : \mathcal{U} \to \mathbb{X}$ is bijective onto U. Since $\rho(s_i)$ is an isometry for the Riemannian metric in $\mathbb{X}$, we get that $\mathrm{Tess}(\rho)$ inherits a Riemannian metric from the symmetric space.

The tessellated space comes equipped with a W_3-action by isometries by taking $h[(g,x)] = [(hg,x)]$. Moreover, this action admits a fundamental domain of the form $\{[(id,x)] : x \in \overline{\mathcal{D}}\}$ that we will denote as $\widetilde{\mathcal{D}}$. This action makes Dev a $\rho-$equivariant map, and clearly $\mathrm{Dev}(\widetilde{\mathcal{D}}) = \overline{\mathcal{D}}$.

The first step to prove Theorem 4.2 will be to prove the following proposition:

Proposition 4.3. *The map* $\mathrm{Dev} : \mathrm{Tess}(\rho) \to \mathbb{X}$ *is a globally injective local isometry.*

Proof. Local isometry follows from the construction of the riemannian metric in $\mathrm{Tess}(\rho)$, thus the bulk of the argument is to show injectivity.

Notice that using equivariance of the developing map, it is enough to verify that for any non-trivial $g \in W_3$, we get $\rho(g)\mathcal{D} \cap \mathcal{D} = \emptyset$. We will do so using a ping-pong argument adapted to the reflection group. More precisely, let $\Delta_1, \Delta_2, \Delta_3$ be complementary regions of $\bigcup_{i=1}^3 \mathcal{W}_i$ as in Lemma 3.4. By construction, these sets satisfy

$$\rho(s_i)\Delta_j \subset \Delta_i, \ \forall i \neq j. \tag{4}$$

Notice that a reduced representative for g with respect to s_1, s_2, s_3 corresponds to strings of digits $1, 2, 3$, where no identical digits appear consecutively. Equation 4 and an induction argument on the word length shows that if the reduced form of $g \in W_3$ is of the form $g = s_i g'$, then $\rho(g)\mathcal{D} \subset \Delta_i$, implying $\rho(g)\mathcal{D} \cap \mathcal{D} = \emptyset$.

□

Thanks to this proposition, it suffices to verify that $\mathrm{Tess}(\rho)$ is a *complete metric space* in order to prove Theorem 4.2, as a locally isometric injective map between complete Riemannian manifolds is an isometry.

4.3. **Proof of Theorem 4.2.** Now we are ready to prove the main theorem. Because of Proposition 4.3, it is enough to show that $\mathrm{Tess}(\rho)$ is geodesically complete.

Lemma 4.4. *Let* $\widetilde{\gamma} : [0,T) \to \mathrm{Tess}(\rho)$ *be a unit-speed geodesic whose future cannot be extended. Then, it intersects infinitely many translates of the fundamental domain* $\widetilde{\mathcal{D}}$ *for the* W_3*-action.*

Proof. Suppose by contradiction that $\widetilde{\gamma}$ only intersects finitely many translates of $\widetilde{\mathcal{D}}$. Then $\widetilde{\gamma}$ is necessarily contained in $\bigcup_{|g|\leq N} g\widetilde{\mathcal{D}}$ for N sufficiently large so that all walls intersecting $\widetilde{\gamma}$ are interior to this set. Up to taking a bigger N, we get that the closure of $\mathrm{Dev}(\widetilde{\gamma})$ is contained in the interior of the open set $\mathrm{Dev}(\bigcup_{|g|\leq N} g\widetilde{\mathcal{D}})$. Since $\gamma = \mathrm{Dev} \circ \widetilde{\gamma}$ is a geodesic of $\mathbb{X}$, and $\mathbb{X}$ is a complete metric space, γ can be extended. A small extension of γ beyond T will still be included in $\mathrm{Dev}(\bigcup_{|g|\leq N} g\widetilde{\mathcal{D}})$.

The fact that Dev is an isometry onto its image means that $\widetilde{\gamma}$ can be extended, which is a contradiction. □

This reduces the problem to understanding how geodesics can intersect infinitely many walls in finite time. We will use Corollary 3.6 to understand this .

Proof of Theorem 4.2. Suppose by contradiction that there exists a geodesic that cannot be extended $\widetilde{\gamma} : [0,T) \to \mathrm{Tess}(\rho)$. Observe that $\gamma = \mathrm{Dev} \circ \widetilde{\gamma}$ can be extended in the symmetric space, and $\gamma(T) \in \partial\,\mathrm{Dev}(\mathrm{Tess}(\rho))$.

Let (t'_k) be the increasing sequence of times for which $\gamma(t'_k)$ intersects some translate of a wall $\mathcal{W}_{i'_k}$. We will call the label i'_k the *index* of the wall.

Observe that γ intersects two different walls with a fixed index, it must necessarily intersect a wall with a different index in the middle. This shows that whenever $i'_k = i'_{k+1}$, then the geodesic intersects the same translate of $\mathcal{W}_{i'_k}$ in subsequent times.

To remedy this phenomenon, we define another sequence (t_k) inductively: set $t_1 = t'_1$ and, for $k > 1$,

$$t_k = \min\{t'_j : t_{k-1} < t'_j, \text{the index of the } t'_j\text{-wall differs from the } t_{k-1}\text{-wall}\}.$$

Notice that Lemma 4.4 ensures that this sequence is infinite. Moreover, we see that $t_k \to T$, as otherwise there would be $g \in W_3$ with $\gamma(t) \in \rho(g)\mathcal{D}$ for every t close to T. Any geodesic with closure in a given tile $\rho(g)\mathcal{D}$ can be extended, which would contradict the inextendability.

We will now introduce some notation:

- i_k is the index of the wall containing $\gamma(t_k)$.
- $g_k \in W_3$ is the element for which $\gamma((t_{k+1} - \varepsilon, t_{k+1})) \subseteq g_k\mathcal{D}$, for some $\varepsilon > 0$.
- $P^{i_k}_{r_k}$ for $r_k \in \mathbb{R}$ is the parametrized parallel set of $\mathcal{W}_{i_k}$ whose translate contains $\gamma(t_k)$.

Notice that $g_{k+1} = g_k s_{i_{k+1}}$ by construction. Moreover, Lemma 4.4 ensures that g_k is unbounded.

Observe that Corollary 3.6 (and Remark 3.7) implies the existence of $\delta = \delta(\tau) > 0$ so that:

$$d(P^i_t, P^j_s) < \delta \Rightarrow \begin{cases} j = i+1 \text{ and } t < 0, s > 0, \\ j = i-1 \text{ and } t > 0, s < 0 \end{cases}.$$

Let k_0 be such that for any $k \geq k_0$, we get $d_{\mathbb{X}}(\gamma(t_k), \gamma(t_{k+1})) = |t_k - t_{k+1}| < \delta$. Observe that:

$$d_{\mathbb{X}}(P^{i_k}_{r_k}, P^{i_{k+1}}_{r_{k+1}}) \leq d_{\mathbb{X}}(\gamma(t_k), \gamma(t_{k+1})) = |t_k - t_{k+1}| < \delta.$$

Therefore, our choice of k implies:

- If $r_k < 0$, then $i_{k+1} = i_k + 1$ and $r_{k+1} > 0$.
- If $r_k > 0$, then $i_{k+1} = i_k - 1$ and $r_{k+1} < 0$.

Using both conditions simultaneously, we get that

$$i_{k_0+j} = \begin{cases} i_{k_0}, \ j \text{ even} \\ i_{k_0} \pm 1, \ j \text{ odd}. \end{cases}$$

The sign in the previous formula is determined by the sign of r_{k_0}. In particular, we get that there exists $g' \in W_3$ so that

$$g_{k_0+2j} = g'(s_{i_{k_0}} s_{i_{k_0} \pm 1})^j,$$

where we get a $+$ if $r_{k_0} < 0$, and a $-$ otherwise. We will assume from now on that $r_{k_0} < 0$, though the other case can be checked in the same way.

We started the proof by assuming that $\gamma(T) \in \partial \operatorname{Dev}(\operatorname{Tess}(\rho))$. Up to composing with $\rho(g')^{-1}$, the previous discussion shows that there is a sequence of parallel sets $P^i_{r_k}$, for $i = i_{k_0}$ contained in $\mathcal{W}_i$, and points $x_k \in P^i_{r_k}$ such that $\rho(s_i s_{i+1})^k(x_k) \to x \in \mathbb{X}$. We will see that this produces a contradiction.

Let $(p_k, \ell_k) \in \mathcal{N}$ such that $P(p_k, \ell_k) = \rho(s_i s_{i+1})^k P^i_{r_k}$. Because of Lemma **??**, (p_k, ℓ_k) has a convergent subsequence in $\mathcal{N}$, but because of Lemma 2.18, we have that (p_k, ℓ_k) converges to $F_{i-1} \in \mathcal{F}$, which is a contradiction. □

5. Application to critical exponents

In this section, ρ_τ denotes the representation determined by a positive triple of flags (F_1, F_2, F_3) with triple ratio τ. We will use the fundamental domains to compute asymptotics for the critical exponent $h_H(\rho_\tau)$ of ρ_τ associated with the Hilbert metric when $\tau \to \infty$.

5.1. **Finsler distances and critical exponents.** From , $\rho_\tau : W_3 \to SL_3(\mathbb{R})$ preserves a convex domain $\Omega_\tau \subseteq \mathbb{RP}^2$. Given a properly convex domain, we can define a Finsler metric in it: given $x, y \in \Omega$, and $a, b \in \partial\Omega_\tau$ the limiting points of the segment inside Ω_τ joining them, we define

$$d_H(x, y) = \frac{1}{2} \log[a, x, y, b],$$

where $[x_1, x_2, x_3, x_4]$ is the cross-ratio between four aligned points in $\mathbb{RP}^2$ under the normalization $[0, 1, t, \infty] = t$.

Fixing a basepoint $o \in \Omega_\tau$, we define the *Hilbert critical exponent* as

$$h_H(\rho_\tau) = \lim_{T\to\infty} \frac{1}{T} \log \#\{g \in W_3 : d_H(o, \rho_\tau(g)o) \leq T\}.$$

Note that this quantity is independent of o.

Let $\mathfrak{a} = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\}$, and $\mathfrak{a}^+ = \{(x, y, z) \in \mathfrak{a} : x \geq y \geq z\}$. Fix $q \in \mathbb{X}$. We can define a *Cartan projection* $\kappa : G \to \mathfrak{a}^+$ by defining $\kappa(g)$ as the ordered logarithms of the singular values of g with respect to q.

Notice that the first singular value of g with respect to q is always $\log \|g\|$, and the last one is $-\log \|g^{-1}\|$, where the norms are taken with respect to q.

We will define the *Hilbert functional* $\alpha^H \in \mathfrak{a}^*$ to be $\alpha^H(x, y, z) = \frac{1}{2}(x - z)$. Particularly, notice that $\alpha^H(\kappa(g)) = \frac{1}{2} \log \|g\| \|g^{-1}\|$.

Proposition 10.1 of [Dan+24] says that there exists a $C > 0$ such that

$$|d_H(o, g \cdot o) - \alpha^H(\kappa(g)) \leq C,$$

for all $g \in Aut(\Omega_\tau)$. Thus, we have that

$$h_H(\rho_\tau) = \lim_{T\to\infty} \frac{1}{T} \log \#\{g \in W_3 : \alpha^H(\rho_\tau(g)) < T\}.$$

The goal of this section is to use the geometry of the fundamental domain to prove the following theorem:

Theorem 5.1. *We have that*

$$\lim_{\tau\to+\infty} h_H(\rho_\tau) = \frac{1}{2}$$

Note that $\rho_\tau(s_i s_j)$ is unipotent. As we will see later in the section, this implies that $h_H(\rho_\tau) \geq 1/2$ for every τ, so this is evidence of the degeneration of the representation.

5.2. **Chains of parallel sets.** Let P_S be the parallel set invariant under the order-three element permuting the flags defining ρ_τ. The goal of this section is to estimate the growth of $d_{\mathbb{X}}(P_S, \rho_\tau(g)P_S)$ for $g \in W_3 - \{e\}$. This number is always positive, since P_S and $\rho_\tau(g)P_S$ lie in different translates of the fundamental domain $\mathcal{D}$.

Using Theorem 4.2, we can control $d_{\mathbb{X}}(P_S, \rho_\tau(g)P_S)$ by looking at which parallel sets of walls a minimizing geodesic from P_S to $\rho_\tau(g)P_S$ must cut.

Definition 5.2.

- A *chain* from P_S to $\rho_\tau(g)P_S$ is a sequence $P_1, \dots, P_n$, such that: P_1 is contained in a wall of $\mathcal{D}$, P_n is contained in a wall of $\rho_\tau(g)\mathcal{D}$, and for every $1 \leq k \leq n$, P_k is a parallel set in some translate of the wall, where every consecutive pair P_k, P_{k+1} lies in different walls of a common translate of $\mathcal{D}$. We denote by $\mathcal{C}_\tau(g)$ the set of chains from P_S to $\rho_\tau(g)P_S$.
- The *chain translation length* $D_\tau(g)$ from P_S to $\rho_\tau(g)P_S$ is defined as:

$$D_\tau(g) = \inf_{\{P_i\}} d_{\mathbb{X}}(P_S, P_1) + \sum_{i=1}^{n-1} d_{\mathbb{X}}(P_i, P_{i+1}) + d_{\mathbb{X}}(\rho_\tau(g)P_S, P_n), \tag{5}$$

  where the infimum is taking over chains from P_S to $\rho_\tau(g)P_S$. A chain is *minimizing* if it realizes the infimum.

Our motivation for defining $D_\tau(g)$ comes from the following lemma.

Lemma 5.3. *Given $g \in W_3$ and $\tau \in \mathbb{R}^+$, then*

$$D_\tau(g) \leq d_{\mathbb{X}}(P_S, \rho_\tau(g)P_S).$$

When $\tau = 1$, then equality is achieved, and $D_1(g) = d_{\mathbb{H}^2}(z_B, gz_B)$, where we think of $W_3 \leq \mathrm{Isom}(\mathbb{H}^2)$, and z_B the barycenter of the ideal triangle defining the reflection group.

Proof. For the first part, take a geodesic γ realizing $d_{\mathbb{X}}(P_S, \rho_\tau(g)P_S)$. By Remark 2.14, this geodesic intersects translates of the wall at times $0 < t'_1 < \ldots < t'_k < d_{\mathbb{X}}(P_S, \rho_\tau(g)P_S)$. Dropping consecutive times when the geodesic intersects the same wall, we assume that $\gamma(t_i)$ and $\gamma(t_{i+1})$ lie on different walls of a common tile. Let P_i be the parallel set containing $\gamma(t_i)$, and declare $P_0 = P_S$, $P_{k+1} = \rho_\tau(g)P_S$, then

$$\sum_{i=0}^{k} d_{\mathbb{X}}(P_i, P_{i+1}) \leq \sum_{i=0}^{k} d(\gamma(t_i), \gamma(t_{i+1})) = d_{\mathbb{X}}(P_S, \rho_\tau(g)P_S).$$

The sequences of parallel sets we just constructed form a chain, which finishes the first inequality.

For the second part, note that when the triple ratio is one, all the parallel sets are orthogonal to the same irreducible slice. In particular, the distances between the parallel sets coincide with the distances of their respective projections to the slice. □

Notice that any term in the sum in Equation (5), apart from the first and last, can be computed using Proposition 3.5, and the other two can be computed with Lemma 3.8. In particular, the dependence of the left-hand-side on the triple-ratio is explicit.

Remark 5.4. Looking at the expression for $CR(P_t^i, P_{t'}^{i+1})$ given by Proposition 3.5 and defining $h = 2t + \frac{1}{2}\log\tau$ and $h' = 2t' + \frac{1}{2}\log\tau$, we can rewrite it as:

$$\frac{\left(1 + e^{2t} + \tau e^{2t'+2t}\right)\left(1 + \tau e^{2t} + \tau e^{2t'+2t}\right)}{4\tau e^{2t'+2t}} = \frac{\left(1 + e^{h+h'} + \tau^{-1/2} e^{h}\right)\left(1 + e^{h+h'} + \tau^{1/2} e^{h}\right)}{4e^{h+h'}},$$

One can verify that, for fixed h and h', the right-hand side is a monotonically increasing function of τ for $\tau \in [1, \infty)$.

This suggests that our choice of center for the wall geodesic in the previous section was arbitrary. It is better to shift the center of the geodesics by the factor $\frac{1}{4}\log\tau$, as $P_{-\frac{1}{4}\log\tau}$ represents more clearly what would be the foot of the altitude, as can be seen in Lemma 3.8.

Given $\{P_i\}_{i=1,\dots,n} \in \mathcal{C}_\tau(g)$, we will define sequences $(g_i)_{i=1,\dots,n-1} \in W_3^{n-1}$, $\{j_i\}_{i=1,\dots,n} \in (\mathbb{Z}/3\mathbb{Z})^n$ and $(t_i)_{i=1,\dots,n} \in \mathbb{R}^n$ such that

- P_i and P_{i+1} are adjacent to $\rho(g_i)\mathcal{D}$ for every i.
- P_i is the $\rho(g_i)$ translate of the parallel set $P_{t_i}^{j_i}$ contained in the wall $\mathcal{W}^{j_i}$.

By construction of a chain, we get that $g_i^{-1} g_{i+1} = s_{j_{i+1}}$ for every $i \in \{1, \dots, n-2\}$. Moreover, $g_1 = s_{j_1}$ and $g_{n-1} s_{j_n} = g$.

Conversely, given $(g_i)_{i=1,\dots,n-1}$ verifying the conditions of the previous paragraph, and $(t_i)_{i=1,\dots,n}$ a sequence of real numbers, we can define a unique chain P_i by defining $P_i = \rho(g_i) P_{t_i}^{j_i}$, for j_i the index of the involution $g_{i-1}^{-1} g_i = s_{j_i}$. Notice that the data (g_i), (t_i) is blind to the τ-parameter, so we can construct bijections between parallel chains across different parameters.

More concretely, we define a bijection

$$c_{\tau,\tau'} : \mathcal{C}_\tau(g) \to \mathcal{C}_{\tau'}(g), \tag{6}$$

mapping a chain between P_S and $\rho_\tau(g)P_S$ corresponding to (g_i) and times (t_i), to a chain from P_S to $\rho_{\tau'}(g)P_S$ associated with (g_i), and times (t_i'), where we declare that

$$t_i' = t_i + \frac{1}{4}\log\frac{\tau}{\tau'}, \forall i \in \{1, \dots, n\}.$$

This choice of times is motivated by Remark 5.4, notice the following.

Remark 5.5. Given $(P_i) \in \mathcal{C}_\tau(g)$, we get that $(P_i') = c_{\tau,\tau'}(P_i)$ is a sequence verifying that $d(P_i', P_{i+1}') < d(P_i, P_{i+1})$ whenever $\tau' < \tau$ (and the reverse inequality if $\tau' > \tau$), for $\tau, \tau' \geq 1$.

This leads us to the monotonicity result we describe in the next section.

5.3. **Monotonicity of the chain translation length.** The following fact will be crucial for the critical exponent computation:

Proposition 5.6. *Given $g \in W_3$, then the function $\tau \to D_\tau(g)$ is monotinically increasing in $[1, \infty)$. More precisely, we get the inequality*

$$D_\tau(g) - D_{\tau'}(g) \geq \log\frac{\cosh(\log\tau^{1/6})}{\cosh(\log\tau'^{1/6})},$$

for any $1 \leq \tau' < \tau$.

Proof. Let $\{P_k\}$ be a minimizing chain for P_S and $\rho_\tau(g)P_S$, and let $c_{\tau,\tau'}(P_i) = (P'_k)$ the corresponding chain between P_S and $\rho_{\tau'}(g)P_S$, as constructed in the previous section. By setting $P_0 = P_S$ and $P_{n+1} = \rho_\tau(g)(P_S)$, we get

$$D_\tau(g) - D_{\tau'}(g) \geq \sum_{k=0}^{n} d(P_k, P_{k+1}) - d(P'_k, P'_{k+1}).$$

Remark 5.4 claims that every term, except the first and last, is strictly positive. Thus, it suffices to bound the first and last terms.

Notice that for our choice of t_1 and t'_1, Lemma 3.8 implies

$$\frac{CR(P_S, P_1)}{CR(P_S, P'_1)} \geq \frac{\cosh(\log \tau^{1/6})}{\cosh(\log \tau'^{1/6})}.$$

Since

$$\frac{CR(P_S, P_1)}{CR(P_S, P'_1)} \leq e^{2(d_{\mathbb{X}}(P_S,P_1) - d_{\mathbb{X}}(P_S,P'_1))},$$

taking logs and repeating the argument for the other end finishes the proof. □

This suggests that critical exponents in the symmetric space, or, as we will later see, the Hilbert metric, should decrease monotonically in τ. However, the additive constant between $D_\tau(g)$ and $D_{\tau'}(g)$ is not enough to obtain this result.

5.4. **The Farey lamination and bounds in terms of the Coxeter length.** We can think of W_3 as the subgroup of $\mathrm{Isom}(\mathbb{H}^2)$ generated along edges of the ideal triangle Δ_0 with vertices at 0,1,∞. The orbit of this triangle is called the *Farey lamination*.

Observe that the equivariance of the walls and the existence of the fundamental domain (Theorem 4.2), gives a W_3-equivariant map f_τ between the orbit of the walls by the action of the group and the Farey lamination in $\mathbb{H}^2$, similar to Farey patterns from [Sch25]. We now introduce a variation of the notion of *cutting sequence* associated to $g \in W_3$, introduced in Section 2 of [Ser85].

Again we think of W_3 as the subgroup fixing the ideal triangle with vertex 0, 1 and ∞. Let $i \in \mathbb{H}^2$, $g \in W_3$ and take γ be the oriented geodesic in $\mathbb{H}^2$ from i and gi. Observe that this geodesic intersects different leaves of the lamination at points $\gamma(0), \gamma(t_1), \ldots, \gamma(t_n)$, where $\gamma(t_n) = gi$. For each arc from $\gamma(t_i)$ to $\gamma(t_{i+1})$, the two leaves of the lamination containing $\gamma(t_i)$ $\gamma(t_{i+1})$ intersect at an ideal point in $\partial\mathbb{H}^2$. We associate the symbol L or R according to whether the point lies at the left or right of γ. This allows us to associate to g, a *cutting sequence* $L^{k_1}R^{k_2}\ldots L^{k_j}$, where each $k_i \in \mathbb{N}^+$, except possibly k_1 and k_j (that may be zero) (see Figure 5.4).

Definition 5.7. Given $g \in W_3$ with cutting sequence $L^{k_1}R^{k_2}\ldots L^{k_j}$, we will call each k_i the *depth of a cusp excursion*. We will call the number $C(g)$ of alternations between powers of L and powers of R the *Coxeter length* of g. A visualization of a cutting sequence can be seen in Figure 5.4

Remark 5.8. Consider the Farey pattern generated by the ideal triangle joining the vertex 0, 1 and ∞ of $\partial\mathbb{H}^2$. The stabilizer of this pattern in $\mathbb{H}^2$ is, up to index two, $SL_2(\mathbb{Z})$, and up to index 6, W_3. Moreover, the only element of this stabilizer fixing $\{\Re z = 0\}$ is the reflection along this geodesic.

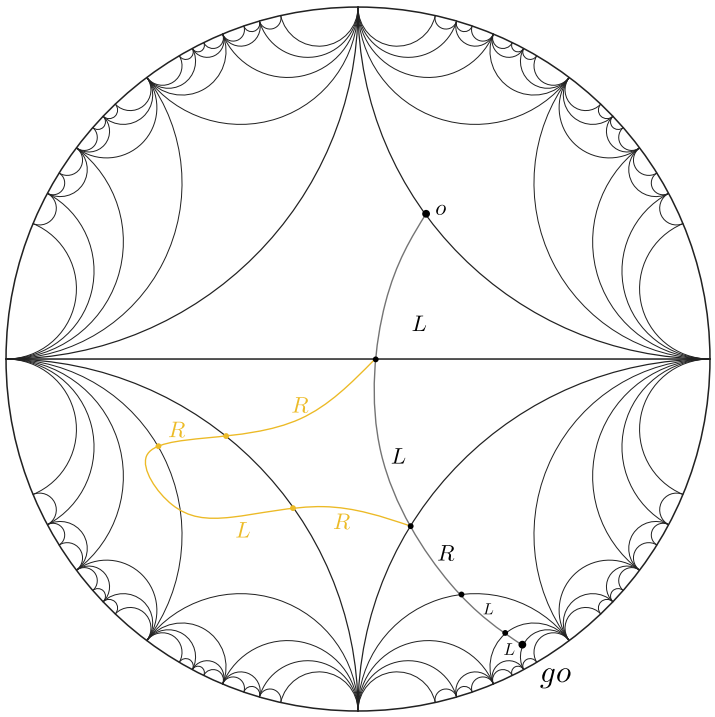


FIGURE 7. A representation of the argument of the Lemma 5.11. Any deviation from the geodesic path must contain at least one turn to the left, thus increasing the number $C(g)$. In this case, $C(g) = 2$, as there are 2 turns from left to right.

Setting

$$L = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \; R = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

Associates a cutting sequence with an element of the stabilizer of the Farey lamination. The importance of this comes from the following lemma:

Lemma 5.9. *Let $g \in W_3$ with cutting sequence $s_1 \dots s_n = L^{k_1} R^{k_2} \dots L^{k_j}$, and let $M = L^{k_1} R^{k_2} \dots L^{k_j} \in PSL_2(\mathbb{Z})$. If $\Re(gi) > 0$, then $gi = Mi$. In general, $d(i, gi) = d(i, Mi)$.*

Proof. The rotation $S(z) = -1/z$ fixes i, preserves the Farey pattern and preserves orientation, so the segment from i to Sgi is the S-image of the one from i to gi and has the same cutting sequence. Hence we may assume $\Re(gi) > 0$.

If $C(g) = L^{k_1} \dots L^{k_j}$, we denote by M the product of the matrices representing L and R. For $k > 0$, denote M_k the product of the first k elements of $C(g)$. We will prove by induction that $C(M) = C(g)$.

Let γ be the geodesic arc joining i to gi. Let $e_0 = \{\Re z = 0\}$ and let T_0 be the ideal triangle with vertices 0, 1 and ∞. Let $e_0, e_1, \dots, e_n$ be the edges of the Farey tessellation met by γ, in order, and for $0 \leq k < n$ let T_k be the triangle crossed

by γ between e_k and e_{k+1}. Since $\Re(gi) > 0$, the segment γ leaves e_0 into T_0, and since gi lies on an edge, it lies on e_n. The letter u_{k+1} records whether γ leaves T_k through its left or its right edge, as seen when entering T_k through e_k.

We prove by induction on k that $e_k = M_k e_0$, and $T_k = M_k T_0$ when $k < n$. The case $k = 0$ is trivial. Now assume $e_k = M_k e_0$ and $T_k = M_k T_0$. Since M_k preserves the tessellation and the orientation, and maps e_0 to e_k and T_0 to T_k, it maps the left (resp. right) edge of T_0 seen from e_0 to the left (resp. right) edge of T_k seen from e_k, and likewise for the triangles beyond them. Therefore

$$e_{k+1} = M_k u_{k+1} e_0 = M_{k+1} e_0, \qquad T_{k+1} = M_k u_{k+1} T_0 = M_{k+1} T_0.$$

In particular $gi \in e_n = Me_0$, so $h = M^{-1}g$ maps i into e_0. Since h preserves the tessellation and distinct edges have disjoint interiors, $he_0 = e_0$. An element of the stabilizer of the Farey tesselation preserving e_0 either necessarily fixes i, so in particular $gi = Mi$.

□

Remark 5.10. Note that the lemma above also implies that g and M differ up to an element that fixes the Farey tesselation and the geodesic joining 0 with $+\infty$.

Now we will relate $C(g)$ with chains of parallel sets. Let $P_1, \dots P_n$ be a chain of parallel sets joining P_S with $\rho(g)P_S$. Note that each P_i with $i \in \{1, \dots, n\}$ belongs to a wall, and thus we can associate a *cutting sequence* $C(\{P_i\})$ using the bijection $c = c_{\tau,1}$ from Equation (6) earlier in this section (according to whether $c(P_i)$ lies at the left or the right of $c(P_{i+1})$), see Figure 5.4.

Lemma 5.11. *Let $\rho : W_3 \to SL_3(\mathbb{R})$ be a positive representation, and $g \in W_3$, then, for every chain $\{P_i\}$ associated to g, we have that $C(g) \leq C(\{P_i\})$.*

Proof. Note that, because the cutting sequence associated to a chain of flags was defined in terms of a Farey pattern, the proof of the lemma can be restricted to $\mathbb{H}^2$. We denote by T the projection of $\mathcal{D}$ to the invariant totally geodesic $\mathbb{H}^2$.

Let $\{e_i\}_{i \leq m}$ be the sequence of leaves of the Farey tesselation crossed by the geodesic arc γ joining z_B and $\rho(g)z_B$, where z_B is the barycenter of the ideal triangle. Note that each e_i is crossed exactly once. Note that each e_i divides $\mathbb{H}^2$ into two conected components.

Note that g can be written uniquely as a product $g = \sigma_{n_m}\sigma_{n_{m-1}} \dots \sigma_{n_1}$ with $n_i \neq n_{i+1}$. Moreover, each e_i lies in the boundary of $\sigma_{n_i} \dots \sigma_{n_1} T$. We will prove the result by induction on m.

When $g = \sigma_i$, $C(g) = 0$, so the base step is trivial. Now assume that g is as above.

We denote H_i^+ the component containing $\rho(g)z_B$, and H_i^- the one containing z_B. By definition it is true that

$$z_B \in \bigcap_{i \leq m} H_i^-, \qquad \rho^F(g)z_B \in \bigcap_{i \leq m} H_i^+.$$

This implies that every chain needs to go at least once through every e_i. By the induction hypothesis, we get that any chain $\{P_i\}$ associated to $\sigma_{n_{m-1}} \dots \sigma_1$ satisfies the inequality

$$C(\{P_i\}) \geq C(\sigma_{m-1} \dots \sigma_1).$$

Let $T_i = \sigma_{n_i} \dots \sigma_{n_1} T$. Every side e_i of T_i has, according to the orientation of an arc entering T_i through the e_i, a side to the right (which we denote $r(e_i)$), and a side to the left, which we denote $l(e_i)$.

Note that a chain $\{e_i\}$ can be defined inductively by applying either r or l to the previous element.

Note that $|C(g) - C(\sigma_{n_{m-1}} \dots \sigma_{n_1})| \leq 1$. If If $C(g) = C(\sigma_{n_{m-1}} \dots \sigma_{n_1})$ the proposition is proven.

If $C(g) = C(\sigma_{n_{m-1}} \dots \sigma_{n_1}) + 1$, then that means that we make a left turn at the edge adjacent to $\sigma_{n_{m-1}} \dots \sigma_{n_1} T$ and a right turn at the edge adjacent to gT (or the opposite). Thus, the proof of the theorem reduces to the following claim (see also Figure 5.4):

Claim: Let T be the fundamental domain of the action of W_3, with e a side. Let $r(e)$ be the side that is on the right. Then every chain from e to $r(e)$ contains a right turn.

Proof of the claim. Let $\{e_i\}_{i \in \mathbb{N}}$ be a path with only left turns. Note that every edge in this path has an ideal endpoint in common (which is the endpoint shared by e and $l(e)$), thus, this path cannot cross $r(e)$. □

The claim concludes the induction hypothesis, which in turn concludes the proof. □

Now we will use the previous lemma to develop an estimate $D_\tau(g)$ in terms of C(g). Corollary 3.6 bounds the minimal distance between P_t^i and an adjacent wall in terms of the function

$$F_\tau(x) = 2 \cosh^{-1} \left(\frac{\sqrt{1+e^x} + \sqrt{1+\tau e^x}}{2} \right).$$

A computation shows the following.

Lemma 5.12. *Given $\tau > 0$, the function F_τ is a 1-Lipschitz, stricly convex function.*

We get the following corollary:

Corollary 5.13. *Let $P_\cdot^i$ denote a parallel set in $\mathcal{W}^i$ at the boundary of our fundamental domain. Then*

$$d(P_s^i, P_t^{i+1}) + d(P_t^{i+1}, \rho_\tau(s_{i+1}) P_r^{i-1}) \geq 2 \cosh^{-1} \left(\frac{\sqrt{1+\tau^{-1/2}} + \sqrt{1+\tau^{1/2}}}{2} \right),$$

for any $s, t, r \in \mathbb{R}$. The same inequality holds for P_s^i, P_t^{i-1} and P_t^{i+1}.

Proof. We know that $d(P_t^{i+1}, \rho_\tau(s_{i+1}) P_r^{i-1}) \geq F_\tau(t)$ by Corollary 3.6. Aplying a duality, we get that $d(P_s^i, P_t^{i+1}) \geq F_\tau(-t - \log \tau)$. Now, by the convexity of the function F_τ, we get

$$d(P_s^i, P_t^{i+1}) + d(P_t^{i+1}, \rho_\tau(s_{i+1}) P_r^{i-1}) \geq F_\tau(t) + F_\tau(-t - \log \tau) \geq 2 F_\tau \left(-\frac{\log \tau}{2} \right),$$

which is the result we wanted. By duality, we obtain a similar result for i, $i-1$, and $i+1$. □

This lemma implies that length of geodesics with one endpoint in $\mathcal{W}^{i+1}$ (resp. $\mathcal{W}^{i-1}$) and another in $\sigma_i\mathcal{W}^{i-1}$ (resp. $\sigma_i\mathcal{W}^{i+1}$) have uniformly bounded length. Moreover, this bound goes to infinity when τ goes to infinity.

Let $\{P_i\}$ be a chain, note that if P_i and P_{i+1} are connected by a right turn (resp. left turn) and P_{i+1} and P_{i+2} are connected by a left turn (resp. right turn), then we are in the condition of the previous paragraph. Thus, we have a proof of the following theorem:

Theorem 5.14. *Let $\tau > 0$, and $g \in W_3$, then*

$$D_\tau(g) \geq C(g)\cosh^{-1}\left(\frac{\sqrt{1+\tau^{-1/2}}+\sqrt{1+\tau^{1/2}}}{2}\right).$$

This estimate is nice for words containing a good proportion of Coxeter positions. However, it is a poor estimate for words that make long excursions around cusps. We balance this by using Theorem 5.14 and Proposition 5.6 simultaneously.

5.5. **Bound on the excursions.** This section gives a lower bound for the translation length in the Fuchsian representation in terms of the cusp excursions, which Proposition 5.6 then uses to bound the chain translation length.

Lemma 5.15. *Let $g \in W_3$ with cutting sequence $g = L^{n_1}R^{n_2}\dots L^{n_k}$, then:*

$$d(i, gi) \geq 2\sum_{j=1}^{k}\log n_j + \log 2.$$

Proof. Up to composing with the involution fixing $(0,\infty)$, we can assume that the segment from i to gi lies to the right of $(0,\infty)$. Notice that by remark 5.8, g agrees with $L^{n_1}R^{n_2}\dots L^{n_k}$ up to multipliying by a reflection along $(0,\infty)$. Particularly, we get that $d(i,gi) = d(i, L^{n_1}R^{n_2}\dots L^{n_k}i)$.

A simple argument by induction shows that each entry of the matrix $L^{n_1}R^{n_2}\dots L^{n_k}$ is $\geq n_1\dots n_k$ (notice that $n_i \geq 1$ for all i except maybe the first and last ones). This shows that:

$$\|g\|_2 \geq 2\prod_{j=1}^{k} n_j,$$

where the norm is taken with respect to the standard inner product, as $SO(2)$ stabilizes i. Recall that $\|g\|_2^2 = 2\cosh(d) \leq 2e^d$. Taking logs and using the estimate above gives us the desired bound. □

As a consequence, one recovers the following classical computation:

Lemma 5.16. *Let $A = \begin{pmatrix}1 & 1\\ 0 & 1\end{pmatrix}$ as in the previous lemma, then the critical exponent $h_{\langle A\rangle} = \frac{1}{2}$.*

Proof. Notice that $d(o, A^n o) \geq 2\log n + \log 2$, particularly, we have around $e^{R/2}$ elements in a ball of radius R. □

The following proposition will be crucial. Informally, it quantifies how many elements with high Coxeter index are in a hyperbolic ball of radius R.

Proposition 5.17. *Let $\zeta(x) = \sum_{n\geq 1}\frac{1}{n^x}$ denote the Riemann zeta function, then for any $y > \frac{1}{2}$ and $x > \log\zeta(2y)$ we get that $\sum_{g\in W_3} e^{-xC(g)-yd(i,gi)}$ converges.*

Proof. It suffices to work with the index-two subgroup F_2. Notice that $g = a^{n_1} \dots b^{n_{r+1}}$ occurs for arbitrary indices, and that $C(g) = r + 1$. This allows us to rewrite:

$$\sum_{g \in W_3} e^{-xC(g) - yd(i,gi)} = \sum_{r \geq 1} e^{-xr} \sum_{C(g)=r} e^{-yd(i,gi)}$$
$$\leq C \sum_{r \geq 1} e^{-xr} \left(\sum_{n_1, \dots, n_{r+1}} (n_1 \dots n_r)^{-2y} \right).$$

Notice that $\sum_{n_1,\dots,n_{r+1}} (n_1 \dots n_r)^{-2y} \leq \sum_{r \geq 1} \xi(2y)^r$. Putting everything together, we obtain a geometric series that converges whenever the proposition's conditions are met. □

5.6. **Critical exponents for the Hilbert length.** In this section, we will reap the estimatives shown in the previous section in order to compute the asymptotics of the critical exponent for the Hilbert length. The following remark will be useful:

Remark 5.18. Whenever P and Q are two parallel sets with $CR(P, Q) > 1$, then $d_{\mathbb{X}}(P, Q)$ can be computed by $CR(P, Q) = \cosh^2(d_{\mathbb{X}}(P, Q)/2)$. Using that $\frac{1}{4}e^x \leq \cosh^2(x/2) \leq e^x$, we obtain that $d_{\mathbb{X}}(P, Q) - \log 4 \leq \log CR(P, Q) \leq d_{\mathbb{X}}(P, Q)$.

Cross ratios are easy to compute, and in fact they allow us to relate $d_{\mathbb{X}}(P_S, \rho(g)P_S)$ to the *Hilbert translation length*.

Lemma 5.19. *Let $q \in P_S$ be an arbitrary choice of basepoint with associated Cartan projection κ, then:*

$$d_{\mathbb{X}}(P_S, \rho(g)P_S) \leq \alpha^H(\kappa(g)) + \log 4.$$

Proof. Let $([v_S], [\alpha_S]) \in Y$ associated to P_S . Suppose v_S is unitary for q, then:

$$CR(P_S, \rho(g)P_S) = \langle g^{-1}v_S, v_S \rangle \langle gv_S, v_S \rangle \leq \|g\|^{-1}\|g\|,$$

where we used that $\ker \alpha_S = \langle v_S \rangle^{\perp_q}$, and the Cauchy-Schwartz inequality. Taking logs and the previous remark shows the desired inequality. □

Proposition 5.20. *When $\tau \to \infty$, the critical exponent of the series $\sum_{g \in W_3} e^{-sD_\tau(g)}$, converges to $\frac{1}{2}$.*

Proof. The critical exponent $\rho_\tau(s_2 s_1)$ is always $\frac{1}{2}$. Thus, we know the critical exponent of the desired series is $\geq \frac{1}{2}$. We now need to show that the series converges for any $s > 1/2$.

We have two bounds for $D_\tau(g)$:

- Theorem 5.14 implies that $D_\tau(g) \geq C(g)G(\tau)$, for a function $G(\tau)$ that goes to infinity when $\tau \to \infty$.
- Proposition 5.6 implies that $D_\tau(g) \geq D_1(g)$, and $D_1(g) = d_{\mathbb{H}^2}(i, gi)$.

Particularly, we get that $D_\tau(g) \geq t d_{\mathbb{H}^2}(i, gi) + (1 - t)C(g)G(\tau)$ for any convex combination $t \in (0, 1)$.

Take $s > \frac{1}{2}$, and consider t such that $st > \frac{1}{2}$ and $t < 1$. We get that:

$$\sum_{g \in W_3} e^{-sD_\tau(g)} \leq \sum_{g \in W_3} e^{-s(1-t)G(\tau)C(g) - std(i,gi)}.$$

Since $G(\tau) \to \infty$ when $\tau \to \infty$, we obtain that $s(1 - t)G(\tau) > \log \xi(2st)$ for sufficiently large τ. Proposition 5.17 tells us that the right-hand side converges.

Thus, the series on the left converges, and s is greater than the critical exponent for sufficiently large τ. Since this holds for any $s > \frac{1}{2}$, we get the desired claim. □

Corollary 5.21. *Let $h(\tau)$ be the critical exponent of the Hilbert metric. Then $\lim_{t\to\infty} h(\tau) = 1/2$.*

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